



# Towards a theory of trade finance<sup>☆</sup>



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## ABSTRACT

Shipping goods internationally is risky and takes time. To allocate risk and to finance the time gap between production and sale, a range of payment contracts is utilized. I study the optimal choice between these payment contracts and their implications for trade. The equilibrium contract is determined by financial market characteristics and contracting environments in both the source and the destination country. Trade increases in enforcement probabilities and decreases in financing costs proportional to the time needed for trade. Empirical results from gravity regressions are in line with the model, highly significant and economically relevant. They suggest that importer finance is as important for trade as exporter finance.

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## 1. Introduction

Shipping goods internationally is risky and takes time. Therefore, trading partners not only have to agree on the quantity and the price of the goods traded, but also on how to share the risk and how to finance the time gap between production and sale. To that end, a range of payment contracts is utilized that allocate risk and financing costs differentially between trading partners. These can be broadly classified into exporter finance (Open Account), importer finance (Cash in Advance) and bank finance (Letter of Credit). While there is considerable heterogeneity in the usage across countries, each of the three contracts finances a substantial part of world trade. Survey evidence in [IMF \(2009\)](#) indicates that Open Account (42%) is the most important

contract, followed by bank intermediated transactions (36%) and Cash in Advance (22%).<sup>1</sup> The total value of trade finance is large. [Auboin \(2009\)](#) estimates that about 90% of trade transactions involve some form of trade finance and that the overall market for trade credit and insurance has a size of about \$10–12 trillion.

In recent years, trade finance has become a major concern of policy makers. In 2009, amid fears that trade would contract due to financing constraints, G20 policy makers agreed to make \$250 billion of trade finance support available over a period of two years.<sup>2</sup> These concerns are in line with recent research that shows that financial conditions in source countries can affect trade flows.<sup>3</sup> While this line of research has established a general link between financial conditions and trade, the focus has not been on the underlying mechanisms.

In particular, central aspects of trade finance, such as the payment contract choice of firms, have not been formally analyzed.

To fill this gap, this paper develops a first model of payment contract choice in international trade, where firms choose between the

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<sup>1</sup> See [Section 2](#) for a discussion of additional evidence.

<sup>2</sup> See [G20 \(2009\)](#).

<sup>3</sup> [Amiti and Weinstein \(2011\)](#) and [Paravisini et al. \(2011\)](#) use firm level export data linked with financial data to establish a causal relationship between financial conditions and trade. Some papers document correlations between financial conditions and international trade patterns at the industry level. See in particular [Beck \(2002\)](#), [Beck \(2003\)](#), and [Manova \(2013\)](#). For work on financial constraints and trade see also [Greenaway et al. \(2007\)](#) and [Muûls \(2008\)](#). Finally, effects of financial crisis on trade flows are studied among others by [Berman and Martin \(2012\)](#), [Bricongne et al. \(2012\)](#), [Chor and Manova \(2012\)](#) and [van der Veer \(2010\)](#).

three predominant contract forms, Cash in Advance, Open Account and Letter of Credit.<sup>4</sup>

The model addresses several key questions: Which trade-offs does a firm face when choosing between different payment contracts? Why does the usage of payment contracts differ across countries? How does trade finance affect trade prices, quantities and patterns?<sup>5</sup>

To answer the first two questions, the paper develops a parsimonious micro model that captures the key trade-offs that firms take into account when financing trade transactions. In a second step, the payment contract choice model is incorporated into a standard intra-industry trade model to derive predictions on aggregate variables of interest. The new prediction that financial conditions in both the source and destination country affect trade is then tested using a panel of bilateral trade data.

In the micro model, one risk-neutral exporter is matched with one risk-neutral importer. The two firms play a one shot game. The exporter makes a take it or leave it offer to the importer, specifying the type of contract, the price to be paid and the quantity to be delivered.

The model features two key problems which need to be resolved by the trading partners. First, the time delay from production in the source country to sales in the destination country gives rise to a financing need. Second, the required advance financing by one party generates a commitment problem on the side of its trading partner.

Consider the financing problem first. As trade takes time, either the working capital (Open Account and Letter of Credit), the prepayment (Cash in Advance) or the Letter of Credit fee has to be financed in advance. Financial markets are assumed to be segmented across countries, so that interest rates can differ between the source and the destination country. The choice of payment contract therefore determines the overall financing costs of the transaction.

The other key aspect is a commitment problem that is present because firms can be of bad type and contract enforcement is imperfect. A bad firm tries to deviate from its contractual obligation whenever this is profitable for it. Under Cash in Advance, the exporter might not deliver the goods after receiving the payment. Under Open Account, the importer might not pay the agreed price for the goods after receiving and selling them. To address this commitment problem, a firm can try to enforce the contract in court, which gives rise, however, to enforcement costs. The probability of enforcing a contract successfully in court depends on the country and is given exogenously.

Payment contracts thus trade-off the financing requirements and the risk of a transaction. Under Cash in Advance, enforcement takes place in the source country whereas financing is done in the destination country. Under Open Account, enforcement has to be assured in the destination country whereas financing takes place in the source country. Under a Letter of Credit, commitment problems are solved, but financing has to be done in both countries and additional bank fees are incurred.

Therefore, to maximize exporter profits, a transaction should, in general, be financed by the firm in the country with the lower financing

costs and the weaker contract enforcement. This minimizes interest rate costs and the probability that the trading partner who did not pre-finance the transaction defaults on her contractual obligations. For similar financing costs, exports to a country with weak enforcement should be done through Cash in Advance, while countries with good enforcement should be supplied through Open Account. When two firms in countries with weak contract enforcement trade with each other, bank finance (Letter of Credit) is most useful as it resolves commitment problems on both sides.

Incorporating the contract choice model into a standard intra-industry trade model delivers several new results. First, the model predicts that both source and destination country financial and legal conditions affect trade. This is in contrast to the previous literature on financial conditions and trade that focused on source country characteristics only. Second, the model implies that trade finance costs are variable trade costs proportional to the value of goods exported, similar to the iceberg trade cost formulation as introduced by Samuelson (1954). Third, the model shows that under any contract, conditions in both the source and the destination country matter for trade. That is either the financing cost or the contract enforcement ability of a trading partner enters expected profits. Thus even firms in a country like the US which has strong contract enforcement and a well-developed financial sector are dependent on institutions of their trading partners. Finally, the payment contract choice model has implications for the behavior of trade during a financial crisis, as the ability to switch between different payment contracts can allow firms to mitigate adverse effects of crisis.<sup>6</sup> While most of this paper focuses on deriving the contract choice model and the first two implications, the other two observations constitute interesting starting points for further research.

In the empirical part of the paper, the predictions of the model on the effects of financial conditions on aggregate trade flows are tested, in particular the role of importer finance. I run gravity regressions including interaction terms between distance and financing costs in the source and the destination country. I find that two countries trade less with each other if their financing costs are higher. As predicted, this effect is the larger, the more time is needed for trade.<sup>7</sup> According to the regressions importer financing matters as much for international trade as exporter financing. This is exactly what the model predicts. Strikingly, the literature so far has focused almost exclusively on source country characteristics.

The paper is related to two strands of theoretical literature. First, there are a large number of papers that study the use of trade credit between firms. Trade credit usually refers to downstream lending between firms in a supply chain, both inside a country and across borders. The literature has concentrated on the relationship of two firms inside a country and has studied under which circumstances trade credit is used as a substitute for bank credit and what the underlying costs and benefits are.<sup>8</sup> In this paper, the focus is instead on the trade-off between financing costs and contracting environments in different countries to optimally finance trade transactions.

Second, there are theoretical papers that have considered the relationship between financial market conditions and international trade.<sup>9</sup> Chaney (2005) develops a theoretical model analyzing financial

<sup>4</sup> Additional instruments for trade finance such as for example factoring or forfaiting exist. Furthermore, trade credit insurance can play a role, in particular to facilitate transactions on Open Account terms. For tractability, the current setup focuses on the three main payment contracts in use. Factoring and forfaiting can best be seen as variations of the Letter of Credit discussed here as they are other instruments for transferring the risk from the two trading parties to third parties. Trade credit insurance is not required in the model as firms are assumed to be risk neutral. For more details on the different types of payment contracts, see U.S. Department of Commerce (2008).

<sup>5</sup> The first question is to some extent discussed informally in the professional literature on trade finance. U.S. Department of Commerce (2008) for example explains how payment contracts allocate risk between exporters and importers and how selling on Open Account terms can make the offer of an exporter more competitive. As stated above, this paper goes beyond the practical literature by formalizing the problem of contract choice. It furthermore shows how the payment contract choice jointly determines the allocation of risk and the financing of a transaction and thus gives rise to a trade-off between contract enforcement capabilities and financing costs across countries.

<sup>6</sup> Suppose, for example, a country experiences a financial crisis that leads to a rise in interest rates. Then, a payment contract switch can limit adverse effects by moving the financing activity to the country of the trading partner. This is not possible if financial conditions in both countries deteriorate at the same time. Therefore, multilateral crises like the recent great recession should have a larger impact on trade flows than national crises.

<sup>7</sup> As no direct time to trade data is available, I use geographical distance as a proxy for the time needed for trade.

<sup>8</sup> See Biais and Gollier (1997), Petersen and Rajan (1997), Wilner (2000), Burkart and Ellingsen (2004), Cunat (2007), Giannetti et al. (2011) and Klapper et al. (2012).

<sup>9</sup> Kletzer and Bardhan (1987) show how sovereign default risk and credit market imperfections can create a comparative advantage. In Matsuyama (2005), the share of pledgeable revenues differs between countries leading to a comparative advantage.

constraints on entry in a heterogeneous firm trade model based on Melitz (2003). [Manova \(forthcoming\)](#) extends this model to a setting where also export volumes can be affected by financial constraints. In [Chaney \(2005\)](#) and [Manova \(forthcoming\)](#) only domestic financial market conditions in the form of financial constraints are relevant for the export decisions of firms. In particular, there is no role for financial market conditions and the contracting environment in the destination country and for the costs of trade finance.

While this is the first paper to formally study the choice of payment contracts for trade finance, some other aspects of trade finance have been discussed in policy papers.<sup>10</sup> Several more recent contributions also study theoretical and empirical aspects of trade finance. [Ahn \(2010\)](#) analyzes the reaction of domestic and international trade finance to financial crisis. [Olsen \(2010\)](#) elaborates on the idea, also discussed in this paper, that enforcement between banks might be easier than between two trading partners as the former interact more frequently. Two papers, [Antràs and Foley \(2011\)](#) and [Glady and Potin \(2011\)](#) directly build on the theory developed here following [Schmidt-Eisenlohr \(2009\)](#). [Antràs and Foley \(2011\)](#) use data from a large US food exporter to test predictions of the model and some extensions. [Glady and Potin \(2011\)](#) focus on the role of letters of credit. Finally, [Engemann et al. \(2011\)](#) and [Eck et al. \(2011\)](#) study the role of international trade credit in addressing problems arising from information asymmetry between exporters and importers.

The empirical part of the paper is closest to the papers that study the relationship between the financial development of a country and the sectoral concentration of its exports.<sup>11</sup> It differs, however, in two central aspects. First, motivated by the theoretical results, I consider financing costs both in the source and the destination country.<sup>12</sup> Second, instead of focusing on sectoral differences, a distance interaction is employed to test for an effect of financial conditions proportional to the time needed for trade.<sup>13</sup>

The rest of the paper is organized as follows. [Section 2](#) discusses survey evidence on payment contract use across countries. [Section 3](#) introduces a model of payment contracts. [Section 4](#) sets the model into a standard intra-industry trade framework and derives implications for trade patterns. [Section 5](#) presents the empirical results. [Section 6](#) concludes.

## 2. The use of payment contracts

The use of payment contracts varies widely across countries. This can be seen in [Table 1](#), which reports data from a survey by the Finance, Credit & International Business Association (FCIB). The FCIB is an association of finance, credit and business executives. In the survey they ask managers which payment contracts are top payment methods used when dealing with a specific country. The survey is run on the internet platform of the FCIB. As participants self-select into the survey, the data are not from a representative sample of firms and response rates cannot be calculated. Despite these limitations, the survey is interesting because, different from [IMF \(2009\)](#), it asks for the relevance of payment contracts by country.

**Table 1**

Payment contract use, FCIB International Credit & Collections Survey. Share of firms that state that Cash in Advance, Open Account or Letter of Credit is a top payment method for transacting with the country in 2010. Only countries with more than 10 answers are included. This is the case for 59 countries.

|          | CIA            |      | OA            |      | LC            |      |
|----------|----------------|------|---------------|------|---------------|------|
| Top 1    | Ukraine        | 58.3 | Denmark       | 92.9 | China         | 32.0 |
| 2        | Kenya          | 50.0 | Finland       | 92.3 | India         | 32.0 |
| 3        | Nigeria        | 45.5 | Norway        | 90.9 | Jordan        | 30.7 |
| Bottom 1 | United Kingdom | 2.5  | Ukraine       | 16.7 | Norway        | 0.0  |
| 2        | Sweden         | 5.6  | Lebanon       | 18.0 | Costa Rica    | 0.0  |
| 3        | Belgium        | 6.3  | Kenya         | 23.0 | Puerto Rico   | 0.0  |
| Mean     | All countries  | 22.2 | All countries | 60.3 | All countries | 10.8 |

[Table 1](#) reports the top and bottom 3 destination countries for each payment contract. The listings indicate that riskier destination countries tend to be served by Cash in Advance whereas trade with safer countries is financed using Open Account. Letters of Credit never represent more than about a third of the responses. This financing form is most popular in China and India.<sup>14</sup>

## 3. Payment contracts

This section develops a microeconomic model of the choice between Open Account (exporter finance), Cash in Advance (importer finance) and Letter of Credit (bank finance). This choice is particularly relevant for international trade for two reasons.

First, the time gap between the production of goods and the realization of sales revenues is longer for international trade than for domestic sales.

As [Hummels and Schaur \(forthcoming\)](#) report, physical transport times can be substantial in international trade, in particular, when goods are transported by sea. Additionally, [Djankov et al. \(2010\)](#) document that formal procedures necessary for international trade transactions can be extensive, implying a delay from the factory gate to the means of transportation as well as at the border of the importer.<sup>15</sup> This implies that working capital requirements for international trade are larger than for domestic sales.

Second, it is more difficult to enforce contracts across borders. This can be due to differences in legal systems or working languages and a limited willingness of governments to enforce international contracts to the same degree as national ones. Whereas domestic sales naturally take place in a common contracting environment, international trade in general does not. Furthermore, in international trade, a firm might not have a permanent representation in the country of the trading partner, making litigation more difficult and costly. Consequently, international trade is more risky and the allocation of risk more important.<sup>16</sup>

### 3.1. Setup

Each exporter in the source country is matched with one importer in the destination country. A firm can be of good and bad type. Let

<sup>10</sup> [Menichini \(2009\)](#) discusses inter-firm trade finance. She suggests that shocks are propagated through credit chains. Furthermore, she argues that the use of trade finance might be restricted if institutions like contract enforcement and bankruptcy laws are not sufficiently developed. [Ellingsen and Vlachos \(2009\)](#) develop a model of trade credit in a liquidity crisis. Evidence on firm level trade finance of African exporters is documented by [Humphrey \(2009\)](#).

<sup>11</sup> See [Beck \(2002, 2003\)](#) and [Manova \(forthcoming\)](#).

<sup>12</sup> [Manova \(forthcoming\)](#) is an exception. It mentions results from a regression with destination country variables. She finds effects about one third of the size as compared to the effects of source country variables. The model in [Manova \(forthcoming\)](#) does, however, not feature any role for destination country conditions.

<sup>13</sup> In contemporaneous empirical work on the financial crisis, [Paravisini et al. \(2011\)](#) include an interaction term between distance and credit supply of a firm but do not find any significant effects.

<sup>14</sup> Additional to Norway, Costa Rica and Puerto Rico, there are 7 more countries where no firm mentions Letter of Credit as a top payment contract. These are Denmark, Dominican Republic, Finland, Ecuador, Guatemala, Panama and Argentina. While the share of Letter of Credit might be low in these countries it is very unlikely than any of them does not use Letters of Credit at all. The zero values result from a combination of measurement problems related to the phrasing of the question and the small number of responses.

<sup>15</sup> [Amiti and Weinstein \(2011\)](#) calculate that these two causes of delay add up to approximately two months for the median case.

<sup>16</sup> Parties can potentially agree on a court in a third country to provide the contract enforcement. Still, even in that case, the ruling would have to be enforced in the country where the value of the transaction is actually allocated, that is any arrangement requires some degree of local enforcement.

$\eta \in (0,1)$  denote the share of good firms in the source country. Good firms always fulfill their contractual obligation, whereas bad firms deviate whenever this is optimal for them.<sup>17</sup> The type of a firm is private information. Firms are risk neutral.<sup>18</sup> The exporter and importer play a one-shot game.<sup>19</sup>

First, the exporter makes a take it or leave it offer to the importer.<sup>20</sup> Then, the exporter can produce and send off goods to the destination country. Sent goods arrive at the importer after  $t$  time units and sales revenues are realized. Denote production costs by  $K$  and revenues by  $R$ .

There are two imperfections in the economy. First, markets to finance international trade transactions are segmented and financial intermediaries across countries differ in their efficiency. As a result the interest rate a firm faces depends on its location.<sup>21</sup> Second, there is limited enforcement of contracts. This is captured by an exogenous country-specific probability that a contract is enforced in the case where a firm does not choose to fulfill it voluntarily.<sup>22</sup> Under Cash in Advance, it is the probability that the exporter is forced to deliver the goods after receiving the payment. Under Open Account, it is the probability that an importer has to pay the agreed price for the goods after receiving and selling them.<sup>23</sup> Enforcement is not costless however. Whenever a contract is enforced successfully, a share  $\delta$  of the revenue is lost due to litigation and other costs associated with the non-cooperative behavior of one of the two trading parties.<sup>24</sup> Finally, let  $\lambda \in (0,1]$  and  $r \geq 0$  denote the enforcement probability and the interest rate in the source country, respectively. Variables for the destination country are marked with asterisks.

<sup>17</sup> Considering two types of firms allows to endogenize the limited value of contract constraint that I assumed in Schmidt-Eisenlohr (2009). See Appendix E for an illustration of this case which makes the model more tractable, but requires the additional limited value constraint. The first paper I am aware of to introduce two types of firms into a setting with an importer–exporter pair is Araujo and Ornelas (2007). The most similar approach regarding this point is taken by Gladys and Potin (2011) who also introduce two types of firms and a pooling case. Eck et al. (2011) also study two types of firms and separating and pooling cases. Different to this paper, they study how trade credit can be used as a signaling device to resolve problems of asymmetric information.

<sup>18</sup> As firms are risk neutral, they do not demand trade insurance in the model. If firms were risk averse, but had access to a perfect insurance market charging fair premia, the results of the model should not change.

<sup>19</sup> One shot transactions are common in international trade. Eaton et al. (2011) find for US–Colombia matched importer–exporter pairs that the probability of survival of newly formed trade pairs is less than 50%. Even pairs that have survived for five years or longer have average separation rates above 40%. In Schmidt-Eisenlohr (2011), I study the case of repeated transactions for a simplified version of the model. Then, in some cases, trigger strategies can be implemented to improve upon the one-shot equilibrium. If, however, the separation rate is sufficiently large and thus gains from future trade are relatively small, cooperation is not sustainable and the main trade-offs studied here remain valid.

<sup>20</sup> Appendix C discusses the case where the importer has all bargaining power. While expressions change, all effects of country characteristics on the payment contract choice go in the same direction as in the case analyzed here.

<sup>21</sup> In reality, loan markets are not perfectly segmented as assumed here. Multinational firms, for example, might be able to raise money in multiple locations. Even some smaller firms can profit from interest rates differentials across countries by borrowing locally in foreign currency as documented by Brown et al. (2011) and Basso et al. (2011). The trade-offs discussed in the model thus apply to the extent to which firms are constrained to access finance through their local financial market.

<sup>22</sup> This captures the reduced form of an enforcement game played between the importer and the exporter, which is affected by the legal institutions of the two countries. This could be extended to a model in which firms choose their legal expenditures to achieve or prevent enforcement. In that case the enforcement probability would change with the value at stake and there would be an explicit role for firm heterogeneity.

<sup>23</sup> For simplicity these two enforcement probabilities are assumed to be equal. It would be an interesting extension to consider an asymmetry here. This could be rationalized by the difference between the in-kind nature of Open Account and the cash nature of Cash in Advance. For a formalization of this argument see Burkart and Ellingsen (2004).

<sup>24</sup> In the baseline model outlined in the following, this cost fully falls on the firm that enforces the contract. In Appendix B, I discuss the case where a contract fine can be included in the contract that falls on the defaulting party. Then, additional incentive conditions have to be taken into account. These imply that under some conditions both firms choose to fulfill the contract voluntarily.

### 3.2. Cash in Advance, Open Account and Letter of Credit

#### 3.2.1. Cash in Advance — overview

Cash in Advance corresponds to a full pre-payment by the importer. That is, before delivery, the importer pays an amount  $C^{CIA}$  to the exporter. Then, the exporter decides whether to deliver the goods. If the exporter is of the good type, she always delivers the goods. If the exporter is of the bad type, she tries to default on the contract. With probability  $\lambda$  she is forced to deliver the goods anyways. The importer, however, loses a share  $\delta$  of revenues due to enforcement costs. With probability  $1 - \lambda$  the exporter successfully defaults on the contract. Note that in the absence of a contract fine, a bad exporter always has an incentive to default on the contract.

A bad exporter has two choices. First, she can demand the same pre-payment as a good exporter so that the importer cannot distinguish between the two types. I refer to this case as *pooling*. Second, she can demand a lower pre-payment, revealing her type. This case is referred to as *separating*. For the bad exporter separating is never optimal as long as good firms also choose Cash in Advance. It implies a lower payment without any additional gain. A good exporter cannot signal her type, because any prepayment acceptable for a good exporter is also acceptable for a bad exporter. Therefore, if both types choose Cash in Advance, the pooling case arises. Finally, I consider the case where good firms do not choose Cash in Advance. I show that under a relatively weak parameter restriction this implies that also bad firms do not choose Cash in Advance. If this restriction is fulfilled, it is therefore sufficient to focus on the pooling case when studying Cash in Advance.

#### 3.2.2. Cash in Advance — pooling case

The exporter maximizes her expected profits subject to the importer participation constraint<sup>25</sup>:

$$\text{Good type : } \max_c E[\Pi_{E,g}^{CIA,p}] = C^{CIA,p} - K, \quad (1)$$

$$\text{Bad type : } \max_c E[\Pi_{E,b}^{CIA,p}] = C^{CIA,p} - \lambda K, \quad (2)$$

$$\begin{aligned} \text{s.t.} \\ E[\Pi_I^{CIA,p}] = \frac{\eta + (1-\eta)\lambda(1-\delta)}{(1+r^*)^t} R - C^{CIA,p} \geq 0, \end{aligned} \quad (3)$$

(participation constraint importer)

$$E[\Pi_{E,g}^{CIA,p}] = C^{CIA,p} - K \geq 0 \quad (4)$$

(participation constraint good exporter).

The participation constraint of the importer requires that her expected profits are non-negative. As the exporter has all negotiation power, the participation constraint of the importer binds under the optimal contract. A necessary condition for the pooling contract is that the participation constraint of good exporters holds. The optimal payment  $C^{CIA,p}$  and optimal expected profits of a good and bad exporter, respectively, are:

$$C^{CIA,p} = \frac{\eta + (1-\eta)\lambda(1-\delta)}{(1+r^*)^t} R, \quad (5)$$

$$\text{Good type : } E[\Pi_{E,g}^{CIA,p}] = \frac{\eta + (1-\eta)\lambda(1-\delta)}{(1+r^*)^t} R - K, \quad (6)$$

$$\text{Bad type : } E[\Pi_{E,b}^{CIA,p}] = \frac{\eta + (1-\eta)\lambda(1-\delta)}{(1+r^*)^t} R - \lambda K. \quad (7)$$

<sup>25</sup> It is assumed that for all cases the exporter and importer discount profits with their local interest rates. To compare profits between CIA and OA they have to be discounted to the same time period.



Despite the fact that there are strictly positive gains from trade under CIA as long as  $\frac{1-(1-\eta)\lambda\delta}{(1+r^*)^t}R \geq K$ , production and delivery only take place with probability  $\eta + (1-\eta)\lambda$ .

### 3.2.3. Cash in Advance – separating case

Suppose now that conditions are such that a good exporter does not choose Cash in Advance. Given the ability to default on the contract, a bad firm might still consider to offer a Cash in Advance contract even though this implies revelation of its type. In this case, the importer understands that she deals with a bad firm and adjusts her expected revenue downwards. Her participation constraint becomes:

$$E[\Pi_{I,g}^{CIA,s}] = \frac{\lambda(1-\delta)}{(1+r^*)^t}R - C^{CIA,s} \geq 0. \quad (8)$$

Thus the optimal pre-payment that makes the participation constraint of the importer bind is:

$$C^{CIA,s} = \frac{\lambda(1-\delta)}{(1+r^*)^t}R. \quad (9)$$

The expected profit of a bad exporter under CIA in the separating case is thus:

$$E[\Pi_{E,b}^{CIA,s}] = \frac{\lambda(1-\delta)}{(1+r^*)^t}R - \lambda K. \quad (10)$$

A sufficient condition for the bad exporter not to choose Cash in Advance is that the good exporter does not choose this payment contract and that the expected profits for a bad exporter under separation are less or equal to the expected profits of a good firm under pooling. That is if:

$$E[\Pi_{E,g}^{CIA,p}] \geq E[\Pi_{E,b}^{CIA,s}]. \quad (11)$$

Plugging in and rearranging, this is the case if:

$$\frac{R}{K} \geq \frac{1-\lambda}{1-\lambda(1-\delta)} \frac{(1+r^*)^t}{\eta}. \quad (12)$$

For condition (12) to hold, revenues have to be sufficiently larger than production costs. Remember that the advantage of the bad firm is to save on production costs with some probability. For the rest of the paper, I assume that condition (12) holds. That is, if a good firm does not choose Cash in Advance, it is not profitable for a bad firm to choose Cash in Advance either.<sup>26</sup>

### 3.2.4. Open Account – overview

Open Account represents full payment after delivery. That is, first, the exporter produces the goods and delivers them to the importer. Then, after  $t$  time units, the goods arrive at the importer who sells them. The importer now decides whether to pay the claim of the exporter. If of good type, the importer always pays  $C^{OA}$ . If of bad type, she tries to deviate, but is forced to pay with probability  $\lambda^*$ , giving rise to enforcement losses of  $\delta R$ . Now the exporter can choose between a pooling and a separating strategy. That is she can either ask for a low payment after delivery which good and bad importers accept or she can ask for a higher payment that only bad importers are willing to promise. In the following, I first study the pooling case and then proceed to the separating case. Finally, a condition is derived under which pooling is preferred by the exporter.

### 3.2.5. Open Account – pooling case

For pooling to take place, the participation constraint of a good importer has to be respected:

$$\max_C E[\Pi_E^{OA,p}] = \frac{\eta^* + (1-\eta^*)\lambda^*(1-\delta)}{(1+r)^t} C^{OA,p} - K, \quad (13)$$

$$\begin{aligned} \text{s.t.} \\ E[\Pi_{I,g}^{OA,p}] = R - C^{OA,p} \geq 0 \\ (\text{participation constraint good importer}). \end{aligned} \quad (14)$$

It is optimal for the exporter to choose  $C^{OA,p}$  such that the participation constraint of the good importer binds. This implies:

$$C^{OA,p} = R, \quad (15)$$

$$E[\Pi_E^{OA,p}] = \frac{\eta^* + (1-\eta^*)\lambda^*(1-\delta)}{(1+r)^t} R - K. \quad (16)$$

### 3.2.6. Open Account – separating case

The separating case implies the following participation constraint of a bad importer:

$$E[\Pi_{I,b}^{OA,s}] = \frac{R - \lambda^* C^{OA}}{(1+r^*)^t} \geq 0. \quad (17)$$

It is now optimal for the exporter to make the participation constraint of a bad importer bind. This implies:

$$C^{OA,s} = \frac{R}{\lambda^*}. \quad (18)$$

The prepayment  $C^{OA}$  is chosen such that it exactly offsets the risk of non-payment by the importer. In expectation the importer thus pays  $R$  to the exporter. Note that the expected profits of an exporter are reduced by the fact that only a bad importer accepts the contract. The probability for her to be matched with a bad importer is  $1 - \eta^*$ . Expected profits thus are:

$$E[\Pi_E^{OA,s}] = (1-\eta^*) \left( \frac{(1-\lambda^*\delta)R}{(1+r)^t} - K \right). \quad (19)$$

From a comparison of profits, it is evident, that an exporter prefers a pooling contract to a separating contract if:

$$\frac{R}{K} > \frac{\eta^*(1+r)^t}{\eta^* - (1-\eta^*)(1-\lambda^*)}. \quad (20)$$

A separating contract increases expected payments from bad importers, but reduces the probability of trading as good importers do not accept the offered contract. For the rest of the paper, assume that condition (20) is satisfied and thus exporters always offer a contract that implies pooling.<sup>27</sup>

### 3.2.7. Letter of Credit

A Letter of Credit (LC) captures the case where banks in the source and the destination country are employed to facilitate the trade transaction. The importer pays a fee  $F^{LC}$  to her bank, which issues a letter of credit that guarantees payment to the exporter upon proof of delivery. This fee finances monitoring and other administrative costs related to the issue and execution of a Letter of Credit. Assume that the fee is proportional to the value of the transaction  $C^{LC}$ , that is  $F^{LC} = f^{LC} C^{LC}$ .

<sup>26</sup> To see that the condition is relatively weak, consider the following parameter values:  $\eta = 0.8, \lambda = 0.8, \delta = 0.05, 1 + r^* = 1.1$  and  $t = 0.25$  (3 months). Then, the condition holds if revenues are at least 1.07 times larger than the production costs. This ratio is even smaller if the share of good firms  $\eta$  is higher, if contract enforcement  $\lambda$  is stronger, if the cost of enforcement  $\delta$  is higher, if financing costs  $1 + r^*$  are lower or if the time needed for trade  $t$  is shorter.

<sup>27</sup> This assumption is also relatively weak. Suppose for example that  $\eta^* = 0.8, \lambda^* = 0.8, 1 + r = 1.1$  and  $t = 0.25$  (3 months). Then the condition requires revenues to be at least 1.08 times larger than production costs to rule out the separating contract case. If enforcement in the destination country  $f^{LC} = f^{LC} C^{LC}$  is higher or if the interest rate in the source country  $1 + r$  or the time needed for trade  $t$  is lower, then the required ratio is even lower.

The bank cooperates with a bank in the country of the exporter. Under the assumption of perfect enforcement at the bank level and perfect third party verifiability, this completely resolves the enforcement problem at the individual contract level.<sup>28</sup> With a Letter of Credit, an exporter therefore does not face any risk of non-payment. Her maximization problem is:

$$\max_C E[\Pi_E^{LC}] = \frac{C^{LC}}{(1+r)^t} - K, \quad (21)$$

$$\text{s.t. } E[\Pi_I^{LC}] = \frac{R - C^{LC}}{(1+r^*)^t} - f^{LC} C^{LC} \geq 0 \quad (\text{participation constraint importer}). \quad (22)$$

To maximize her expected profits, the exporter chooses  $C^{LC}$  such that the participation constraint of the importer binds. The optimal payment  $C^{LC}$  and discounted expected exporter profits are:

$$C^{LC} = \frac{R}{1 + f^{LC}(1+r^*)^t}, \quad E[\Pi_E^{LC}] = \frac{R}{(1 + f^{LC}(1+r^*)^t)(1+r)^t} - K. \quad (23)$$

Note that, as pre-financing takes place on both sides, the interest rates of both markets affect profits. As the commitment problems are completely resolved, profits are independent of the enforcement parameters  $\lambda$  and  $\lambda^*$ .

### 3.2.8. Comparison CIA, OA and LC

The six parameters  $r, r^*, \lambda, \lambda^*, \eta, \eta^*$  together with the Letter of Credit fee  $f^{LC}$  and enforcement cost  $\delta$  determine a unique ordering of the different payment forms as stated below. Define  $\tilde{\lambda} = \eta + (1-\eta)\lambda(1-\delta)$  and  $\tilde{\lambda}^* = \eta^* + (1-\eta^*)\lambda^*(1-\delta)$ .

**Proposition 1.** *The optimal choice of payment contract is uniquely determined by the following conditions:*

$$OA \text{ preferred to CIA} \Leftrightarrow \frac{\tilde{\lambda}^*}{(1+r)^t} > \frac{\tilde{\lambda}}{(1+r^*)^t},$$

$$OA \text{ preferred to LC} \Leftrightarrow f^{LC} > \frac{1}{(1+r^*)^t} \left[ \frac{1}{\tilde{\lambda}^*} - 1 \right],$$

$$CIA \text{ preferred to LC} \Leftrightarrow f^{LC} > \frac{1}{(1+r)^t} \left[ \frac{1}{\tilde{\lambda}} - \left( \frac{1+r}{1+r^*} \right)^t \right].$$

**Proof.** See Appendix F.  $\square$

Several predictions, which can be tested with transaction level data, can be derived:

**Corollary 1.** *The usage of*

- Cash in Advance weakly increases in  $r, \lambda, \eta$  and  $f^{LC}$  and weakly decreases in  $\lambda^*$  and  $\eta^*$ . If  $\frac{f^{LC}}{[1 + f^{LC}(1+r^*)^t]^2} < \frac{\tilde{\lambda}(1+r)^t}{(1+r^*)^{2t}}$ , the usage of Cash in Advance also weakly decreases in  $r^*$ .*
- Open Account weakly increases in  $r^*, \lambda^*, \eta^*$  and  $f^{LC}$  and weakly decreases in  $r, \lambda$  and  $\eta$ .*
- Letter of Credit weakly increases in  $\delta$  and weakly decreases in  $r, \lambda, \lambda^*, \eta, \eta^*$  and  $f^{LC}$ .*

**Proof.** See Appendix F.  $\square$

Cash in Advance is more attractive if financing costs and enforcement in the source country are high and if financing costs and enforcement in the destination country are low. Open Account is more profitable if financing costs and enforcement in the destination country are high and if financing costs and enforcement in the source country are low. A Letter of Credit is preferable if financing costs and enforcement in the source country, enforcement in the destination country and Letter of Credit fees are low. Changes in the shares of good firms  $\eta$  and  $\eta^*$  have the same effects on the payment contract choice as changes in the levels of contract enforcement in the respective countries. A higher enforcement cost  $\delta$  makes a Letter of Credit more attractive compared to Open Account or Cash in Advance.

The prediction that Open Account increases in enforcement in the destination country  $\lambda^*$  while Cash in Advance decreases in this variable has been tested recently in Antràs and Foley (2011) using contract level data. Their estimations confirm these two theoretical predictions.

### 3.3. Intermediate contracts

Until now I have only considered contracts with either pre-payment (CIA) or post-payment (OA) as well as Letters of Credit. It is also possible to use what I call an intermediate contract. That is part of the payment is done in advance whereas the remainder is paid after delivery.<sup>29</sup>

Under which circumstances are these contracts preferred to pure importer finance (CIA) or pure exporter finance (OA)? In principle, they could lead to higher expected profits by either saving on financing costs or by reducing commitment problems. In this subsection, I show that, quite surprisingly, intermediate contracts only improve expected profits in one specific case: if, in the absence of intermediate contracts, the exporter chooses OA over CIA and if the interest rate in the destination country is lower than in the source country. In this case, intermediate contracts can be used to reduce financing costs. In any other case, an exporter always chooses either pure exporter finance (OA) or pure importer finance (CIA).

Let  $\phi \in (0,1)$  denote the share of the advance payment in total payment, so the pre-payment at  $t = 0$  is  $C_0 = \phi C$ . In the following, I derive the upper bound for this pre-payment share which makes the exporter indifferent between deviating and always fulfilling the contract.

#### 3.3.1. Resolving the exporter commitment problem

To prevent the exporter from deviating, the payment after delivery has to be sufficiently large. A bad exporter produces and sends the products to the importer if the expected post-delivery payment minus the enforcement costs is larger than the deviation payoff. The latter equals the probability of getting away with cheating  $(1 - \lambda)$  times the production costs saved from doing so  $K$ . The condition therefore is:

$$\frac{\eta^* + (1-\eta^*)\lambda^*}{(1+r)^t} (1-\phi)C - \frac{(1-\eta^*)\lambda^*}{(1+r)^t} \delta R - K \geq -\lambda K. \quad (24)$$

This implies an upper bound on the prepayment share  $\phi$ :

$$\phi \leq 1 - (1+r)^t \frac{1-\lambda}{\eta^* + (1-\eta^*)\lambda^*} \frac{K}{C} - \frac{(1-\eta^*)\lambda^*}{\eta^* + (1-\eta^*)\lambda^*} \delta \frac{R}{C} \equiv \phi^E. \quad (25)$$

When does the exporter prefer an intermediate contract over CIA or OA? Consider the two possible cases. First, suppose that, in

<sup>28</sup> It is conceivable that enforcement is easier between banks than between firms. As banks tend to have more long-term relationships, reputation building and repeated transactions ease enforcement between them. Following this paper, this idea has been looked at in detail by Olsen (2010).

<sup>29</sup> While there is no direct data on the use of intermediate contracts, anecdotal evidence, reported to the author by firms involved in trade finance, suggests that these are also employed in practice.

the absence of intermediate contracts, the exporter chooses OA over CIA. Then, if  $r^* < r$ , an intermediate payment contract can be used to reduce financing costs. This is optimally done up to the point where the incentive constraint of the exporter binds, i.e. where  $\phi = \phi^E = 1 - (1+r)^t \frac{1-\lambda}{\eta^* + (1-\eta^*)\lambda^*} \frac{K}{C} - \frac{(1-\eta^*)\lambda^*}{\eta^* + (1-\eta^*)\lambda^*} \delta_C^R$ . Second, suppose that, in the absence of intermediate contracts, the exporter chooses CIA over OA. Then, if  $r < r^*$ , financing costs could be reduced by introducing some late payment by the importer. Note that the importer commitment problem cannot be resolved, as in the absence of a contract fine, the importer as the last mover always has an incentive to deviate. Suppose some late payment by the importer was desirable. Then, given that financing is cheaper for the exporter and that the importer commitment problem is active independently of the size of the pre-payment, it would be optimal to set the pre-payment to zero. This would correspond to OA, which is a contradiction. Therefore, in this case, an intermediate contract can never be preferred.

To summarize, an intermediate contract can reduce interest rate costs if  $r^* < r$  and OA is preferred in the absence of an intermediate contract. No intermediate contract is chosen if CIA is preferred in its absence.

#### 4. Trade model

What are the implications from payment contracts for quantities, revenues and profits at the firm level and in the aggregate? To address this question, I incorporate the model from the previous section into a standard intra-industry trade model based on [Krugman \(1980\)](#). The analysis delivers new predictions for the patterns of international trade flows and illustrates the similarity between financing costs and iceberg trade costs. Most importantly, the payment contract choice implies that financial and legal conditions in the destination country should matter to the same extent as conditions in the source country.

##### 4.1. Setup

###### 4.1.1. Preferences

There are  $L$  representative consumers in the economy, each supplying inelastically one unit of labor. The individual utility function is:

$$U = \left( \int_{\Omega} q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}. \quad (26)$$

$Q$  is a CES (constant elasticity of substitution) basket of a continuum of differentiated goods. The demand for the differentiated good is:

$$q(\omega) = p(\omega)^{-\sigma} P^{\sigma} Q, \quad (27)$$

where  $\omega$  denotes a variety of the differentiated good,  $P = \left( \int_{\omega \in \Omega} p(\omega)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$  is the price index of the optimal CES basket, and  $\sigma > 1$  is the elasticity of substitution between varieties.

###### 4.1.2. Technology

Labor is the only input factor. Firms operate under monopolistic competition. Each variety is produced by only one firm. There is a fixed cost of production  $f$ . The production of one unit of the differentiated good requires  $a$  units of labor.

##### 4.2. Optimal behavior of firms

Given CES demand and monopolistic competition, firms charge a constant markup over marginal costs to maximize profits. Domestic prices, quantities and profits are:

$$p_d = \frac{\sigma}{\sigma-1} a, \quad q_d = (p_d)^{-\sigma} P^{\sigma} Q, \quad \Pi_d = q_d \left[ \frac{a}{\sigma-1} \right]. \quad (28)$$

Let  $\Pi_x$  denote the profits from exporting. From before, expected exporter profits are:

$$\begin{aligned} E[\Pi_{x,good}^{CIA}] &= \tilde{\lambda} (1+r^*)^{-t} R - K, \\ E[\Pi_{x,bad}^{CIA}] &= \tilde{\lambda} (1+r^*)^{-t} R - \lambda K, \\ E[\Pi_x^{OA}] &= \tilde{\lambda}^* (1+r)^{-t} R - K, \\ E[\Pi_x^{LC}] &= (1+r)^{-t} (1+f^{LC} (1+r^*)^t)^{-1} R - K. \end{aligned}$$

Note that these can be represented by the general expression:

$$E[\Pi_{x,i}] = \alpha^c R - \beta_i^c K, \quad (29)$$

with  $c \in \{CIA, OA, LC\}$  and  $i \in \{good, bad\}$ .

Under Cash in Advance, a bad exporter has lower expected production costs. Thus, she might want to promise a higher quantity at a lower unit price than a good exporter. This is not optimal though as this would reveal her type.<sup>30</sup> Thus, the problem of the good exporter does not only determine the contract type, but also the price and quantity of the goods exported by a firm.

Optimization implies the following export prices, quantities and profits for all contract types  $c$  and firm types  $i$ <sup>31</sup>:

$$p_x^c = \frac{1}{\alpha^c} p_d^*, \quad E[q_{x,i}^c] = A_i^c q_d^*, \quad (30)$$

$$E[R_{x,i}^c] = \frac{1}{\alpha^c} A_i^c R_d^*, \quad E[\Pi_{x,i}^c] = (\alpha^c)^{\sigma} (\sigma - \beta_i^c (\sigma - 1)) \Pi_d^*, \quad (31)$$

with  $A_i^c = (\alpha^c)^{\sigma} \beta_i^c$ .<sup>32</sup>

Note that the parameters  $\alpha^c$  and  $\beta_i^c$ , which represent the costs of trade finance, enter the problem proportional to the value of exports. Thus, in the model, variable trade costs that arise from the financing requirement and the enforcement problem are similar to the iceberg trade cost formulation by [Samuelson \(1954\)](#).

##### 4.3. Implications for trade patterns

Taking into account payment contracts delivers new insights on international trade patterns. In particular, as trade finance can be obtained from financial markets in the source and the destination country, financial conditions and contracting environments in both countries affect trade flows.

In the following, the analysis focuses on the effects of financial conditions on trade flows as this allows for an identification strategy

<sup>30</sup> The condition for the optimality of a pooling contract under Cash in Advance changes from before. This is the case as now the optimal price and therefore  $R$  and  $K$  would differ across types if a separating contract was chosen. Expected profits for a good exporter under pooling are  $\tilde{\lambda}^{\sigma} (1-r^*)^{-\sigma t} \Pi_d^*$  and expected profits of a bad exporter under separation are  $\lambda (1-\delta)^{\sigma} (1+r^*)^{-\sigma t} \Pi_d^*$ . This implies the following new condition for a pooling contract to be the equilibrium outcome:  $\delta > \frac{\lambda^{\frac{\sigma}{\sigma-1}} - (1-\eta)\lambda - \eta}{\lambda^{\frac{\sigma}{\sigma-1}} - (1-\eta)\lambda}$ . This condition is relatively weak. If, for example,  $\eta = 0.8$ ,  $\lambda = 0.8$ , and  $\sigma = 5$ , it is fulfilled for any  $\delta \geq 0$ .

<sup>31</sup> Expected profits of good exporters can be normalized to  $E[\Pi_x] = E[\frac{\Pi_x}{\alpha^c}] = R - \frac{1}{\alpha^c} K$ . Maximizing the original objective function  $E[\Pi]$  implies the same optimal decisions as maximizing the new function  $E[\frac{\Pi}{\alpha^c}]$ . Therefore, the price setting problem is equivalent to the standard case with new per unit production costs of  $\frac{a}{\alpha^c}$ . For details see [Appendix A](#).  
<sup>32</sup>  $E[q_x]$  is the expected quantity, taking into account that under CIA only a fraction  $\lambda$  of export contracts of bad exporters is enforced.

based on bilateral distances. The model implies the following testable predictions:

**Proposition 2.** For given demand conditions  $P^*$  and  $Q^*$  in the destination country, expected export revenues of a good exporter (a bad exporter, keeping the payment contract fixed)

- i) decrease weakly (strictly) if financing costs in the source or (and) the destination country increase:  $\frac{\partial E[R_x]}{\partial(1+r)} \leq 0, \frac{\partial E[R_x]}{\partial(1+r^*)} \leq 0, \frac{\partial E[R_x]}{\partial(1+r)} + \frac{\partial E[R_x]}{\partial(1+r^*)} < 0$ .
- ii) increase weakly if the probability of contract enforcement in the source or the destination country increases:  $\frac{\partial E[R_x]}{\partial \lambda} \geq 0, \frac{\partial E[R_x]}{\partial \lambda^*} \geq 0$ .

**Proof.** See Appendix F.  $\square$

Under any payment contract, the source or the destination country financing costs affect variable costs and thus the volume of trade. Therefore, if financing costs in one country increase, expected export revenues of a firm either decrease or are unaffected. If financing costs in both countries increase, expected export revenues of a firm decrease. Furthermore, expected export revenues of a firm weakly increase in enforcement probabilities in the source and the destination country.

For good firms, the payment contract that maximizes expected profits also implies the highest expected revenues. Thus, in Proposition 2, statements on good exporters hold even when allowing for an endogenous switch of payment contract. Bad exporters imitate the payment contract and quantities of good exporters. Therefore, the statements in Proposition 2 only hold for bad exporters if the payment contract is held fixed.<sup>33</sup>

As discussed before, the working capital requirement of a firm increases in the time needed for trade. Expected export revenues therefore react more strongly to changes in financial conditions if trade takes more time. This is captured in the following proposition:

**Proposition 3.** For given destination country demand conditions  $P^*$  and  $Q^*$ , the log of expected export revenues of a good exporter (a bad exporter, keeping the payment contract fixed)

- i) decreases weakly (strictly) if the log financing costs in the source or (and) the destination country increase:  $\frac{\partial \ln E[R_x]}{\partial \ln(1+r)} \leq 0, \frac{\partial \ln E[R_x]}{\partial \ln(1+r^*)} \leq 0, \frac{\partial \ln E[R_x]}{\partial \ln(1+r)} + \frac{\partial \ln E[R_x]}{\partial \ln(1+r^*)} < 0$ ;
- ii) the more so, the larger the log distance between them:  $\frac{\partial^2 \ln E[R_x]}{\partial \ln(1+r) \partial \ln t} \leq 0, \frac{\partial^2 \ln E[R_x]}{\partial \ln(1+r^*) \partial \ln t} \leq 0, \frac{\partial^2 \ln E[R_x]}{\partial \ln(1+r) \partial \ln t} + \frac{\partial^2 \ln E[R_x]}{\partial \ln(1+r^*) \partial \ln t} < 0$ .

**Proof.** See Appendix F.  $\square$

Proposition 3 predicts that the effect of interest rates on trade flows is increasing in  $\ln t$ , the log of time it takes to transport goods abroad and sell them in the destination country. This provides the theoretical basis for the distance interactions employed in Section 5.

## 5. Empirical tests

The model developed in this paper has predictions both on the payment contract choice at the firm level as well as on aggregate trade flows. While it seems desirable to test both sets of predictions, testing the payment contract choice is difficult due to data limitations.

**Table 2a**

Summary statistics, full sample. This table gives summary statistics for the full sample employed in regressions for Table 5. Variables are: log of bilateral trade value, log of  $(1 + \text{exporter net interest rate margin})$ , log of  $(1 + \text{importer net interest rate margin})$ , log of exporter GDP, log of importer GDP and log of bilateral distance. The table reports the mean, standard deviation, the min and the max of each variable.

| Variable   | Summary statistics N = 142,761 |           |        |        |
|------------|--------------------------------|-----------|--------|--------|
|            | Mean                           | Std. Dev. | Min    | Max    |
| Ln trade   | 8.227                          | 3.71      | −6.925 | 18.875 |
| Ln exp int | .05                            | .034      | .007   | .351   |
| Ln imp int | .05                            | .034      | .007   | .351   |
| Ln GDPE    | 8.902                          | 1.031     | 6.42   | 10.921 |
| Ln GDPI    | 8.848                          | 1.067     | 6.42   | 10.921 |
| Ln dist    | 8.595                          | .857      | 4.088  | 9.901  |

**Table 2b**

Summary statistics, law control sample. This table gives summary statistics for the law control sample employed in the regressions for Table 5. Variables are: log of bilateral trade value, log of  $(1 + \text{exporter net interest rate margin})$ , log of  $(1 + \text{importer net interest rate margin})$ , log of exporter GDP, log of importer GDP and log of bilateral distance. The table reports the mean, standard deviation, the min and the max of each variable.

| Variable   | Summary statistics N = 78,742 |           |        |        |
|------------|-------------------------------|-----------|--------|--------|
|            | Mean                          | Std. Dev. | Min    | Max    |
| Ln trade   | 7.974                         | 3.733     | −6.925 | 18.875 |
| Ln exp int | .05                           | .033      | .009   | .351   |
| Ln imp int | .05                           | .033      | .009   | .351   |
| Exp law    | .185                          | .973      | −1.786 | 1.946  |
| Imp law    | .147                          | .98       | −1.786 | 1.946  |
| Ln GDPE    | 8.913                         | 1.059     | 6.42   | 10.921 |
| Ln GDPI    | 8.851                         | 1.096     | 6.42   | 10.921 |
| Ln dist    | 8.6                           | .846      | 4.088  | 9.901  |

**Table 2c**

Summary statistics, absolute advantage sample. This table gives summary statistics for the absolute advantage sample employed in regressions for Table 9. This is a sub-sample of all country pairs in which one country has an absolute advantage in financing costs ( $\ln(1 + \text{net interest margin})$ ) and the other country has an absolute advantage in contract enforcement (rule of law). Variables are: log of bilateral trade value, log of  $(1 + \text{exporter net interest rate margin})$ , log of  $(1 + \text{importer net interest rate margin})$ , log of exporter GDP, log of importer GDP and log of bilateral distance. The table reports the mean, standard deviation, the min and the max of each variable.

| Variable   | Summary statistics N = 21,119 |           |        |        |
|------------|-------------------------------|-----------|--------|--------|
|            | Mean                          | Std. Dev. | Min    | Max    |
| Ln trade   | 7.961                         | 3.941     | −6.925 | 18.755 |
| Ln exp int | .048                          | .028      | .009   | .278   |
| Ln imp int | .048                          | .027      | .009   | .278   |
| Exp law    | .089                          | .898      | −1.786 | 1.946  |
| Imp law    | .068                          | .904      | −1.786 | 1.946  |
| Ln GDPE    | 8.801                         | 1.029     | 6.42   | 10.921 |
| Ln GDPI    | 8.758                         | 1.063     | 6.42   | 10.921 |
| Ln dist    | 8.542                         | .896      | 4.107  | 9.894  |

In this section, I focus on the two new predictions regarding aggregate trade flows.<sup>34</sup> First, I test whether not only source but also destination market financial conditions affect trade flows. Second, I test whether the effect of financing costs on trade is proportional to the time needed for trade (Proposition 3).

The analysis proceeds in four steps. First, I present the baseline regression that provides evidence for a negative relationship between

<sup>33</sup> Note that switches between payment contracts mostly take place if there are large swings in financing costs or contract enforcement. For smaller changes, exporters keep the same payment contract and only adjust prices and quantities. Furthermore, as, in the model, bad exporters only represent a small fraction of all firms, their effect on aggregate revenues is limited.

<sup>34</sup> This constitutes an indirect test of the payment contract choice model. While the prediction on the destination country effects is new to the theory developed here, it could in principle also be generated by an alternative model that features a role for importer finance. In Appendix D, I derive a test that is more closely related to the payment contract choice as modeled in this paper.



**Table 3**

Financing costs, distance and export volumes. This table analyzes the effects of financing costs in the source and the destination countries and their interactions with distance on export volumes. The dependent variable is the log of exports from country *i* to country *j* in year *t*, 1987–2004. Financing costs are measured by the net interest margin. Time to trade is proxied by the geographical distance between the main cities of two countries. Contract enforcement is proxied by Rule of Law. Regressions in columns 1 and 2 control for the log of GDP per capita, population and GATT status for exporter and importer, respectively. All regressions include a constant and control for a set of bilateral controls as discussed in the text. Column 2 also controls for contract enforcement in both countries. Errors are clustered by exporter–importer pairs. Standard errors are in parenthesis. Significance levels: \*: 10%, \*\*: 5% and \*\*\*: 1%.

| Dependent variable           | Ln bilateral exports |                     |                     |                     |                     |                     |
|------------------------------|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Specification                | (1)                  | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| Ln exp int                   | 39.105***<br>(2.94)  | 12.088***<br>(3.99) |                     |                     |                     |                     |
| Ln imp int                   | 47.159***<br>(3.06)  | 21.649***<br>(3.86) |                     |                     |                     |                     |
| Ln exp int × Ln dist         | −4.783***<br>(0.35)  | −1.670***<br>(0.46) | −5.241***<br>(0.37) | −2.147***<br>(0.49) | −0.469**<br>(0.21)  | −0.082<br>(0.37)    |
| Ln imp int × Ln dist         | −5.752***<br>(0.36)  | −2.654***<br>(0.45) | −6.275***<br>(0.38) | −2.955***<br>(0.48) | −1.432***<br>(0.21) | −2.341***<br>(0.37) |
| Exp law × Ln dist            |                      | 0.220***<br>(0.03)  |                     | 0.215***<br>(0.03)  |                     | −0.049<br>(0.03)    |
| Imp law × Ln dist            |                      | 0.104***<br>(0.03)  |                     | 0.091***<br>(0.03)  |                     | 0.014<br>(0.03)     |
| Ln GDPE × Ln dist            |                      | −0.058**<br>(0.03)  |                     | −0.057**<br>(0.03)  |                     | 0.007<br>(0.06)     |
| Ln GDPI × Ln dist            |                      | 0.164***<br>(0.03)  |                     | 0.173***<br>(0.03)  |                     | 0.119**<br>(0.06)   |
| Ln dist                      | −0.883***<br>(0.03)  | −2.200***<br>(0.29) | −0.844***<br>(0.03) | −2.251***<br>(0.30) |                     |                     |
| R-squared                    | 0.798                | 0.794               | 0.807               | 0.801               | 0.182               | 0.145               |
| N                            | 142,761              | 78,742              | 142,761             | 78,742              | 142,761             | 78,742              |
| # exporter–importer clusters | 18,260               | 17,924              | 18,260              | 17,924              | 18,260              | 17,924              |
| # exporters                  | 144                  | 144                 | 144                 | 144                 | 144                 | 144                 |
| Country controls             | y                    | y                   | –                   | –                   | –                   | –                   |
| Country pair controls        | y                    | y                   | y                   | y                   | y                   | y                   |
| Importer, exporter, year FE  | y                    | y                   | –                   | –                   | –                   | –                   |
| Imp × year, exp × year FE    | n                    | n                   | y                   | y                   | y                   | y                   |
| Country pair FE              | n                    | n                   | n                   | n                   | y                   | y                   |

financing costs both in the source and the destination country and trade flows. I find that, as predicted, the size of the effect of financing costs on trade flows is increasing in the geographical distance between trading partners. Second, I study comparative statics and show that the estimated relationships are economically relevant. Next, I check the robustness of these results. The introduction of interaction terms between geographical distance and measures of contract enforcement (rule of law) and economic development (log of GDP per capita) to the regression does not change the main findings.

Results continue to hold when I introduce exporter × year and importer × year fixed effects and estimate a fixed effects model. Replacing the net interest margin by private capital over GDP as the variable capturing financial conditions delivers very similar results. Finally, I address the question of causality.

### 5.1. Data

I use data on bilateral trade flows from the CEPII trade and production database. The financial market efficiency (net interest margin) and financial market development measures (private credit over GDP) are taken from Beck et al. (2009). The net interest margin is the ratio between the accounting value of the net interest revenues of banks and their total earning assets. It measures the average ex-post markup of the lending activities of banks in a country and therefore represents a measure of financial sector efficiency. This measure differs from ex-ante spreads as it also captures losses on non-performing loans. The alternative measure, private credit over GDP, is a much broader indicator of general financial development. The measure for contract enforcement is extracted from the World Bank Worldwide Governance Indicators. Data on geographical distance and other bilateral indicators is from the CEPII gravity dataset collected by Head et al. (2010). Data on GDP per capita and population are taken from the Penn World Tables (Heston et al. (2009)). The final sample contains 150 exporting countries over the period 1980–2004. When including the net interest rate

margin the number of countries reduces to 144 and the period to 1987–2004. With contract enforcement the years covered are 1998, 2000, and 2002–2004. Descriptive statistics of the main variables for the three data sets used in the main analysis (using the net interest rate margin) are shown in Tables 2a–2c.

### 5.2. Estimation and results

The baseline regression tests the relationship between bilateral trade flows and financing costs (log of  $(1 + \text{net interest rate margin})$ ) in the source and the destination country. It tests both for the direct effect of financing costs and for the effect of their interactions with geographical distance.

$$\ln Y_{ijt} = \zeta_0 + \zeta_1 \ln(1 + r_i) + \zeta_2 \ln(1 + r_j) + \zeta_3 \ln(\text{dist}_{ij}) \cdot \ln(1 + r_i) + \zeta_4 \ln(\text{dist}_{ij}) \cdot \ln(1 + r_j) + \zeta_5 \ln(\text{dist}_{ij}) + \theta'X + \chi_i + \chi_j + \chi_t + \epsilon_{ijt}. \quad (32)$$

An observation  $\ln Y_{ijt}$  is the log trade flow from country *i* to country *j* in year *t*. The regression controls for importer, exporter and year fixed effects and for the log of exporter and importer GDP per capita, exporter and importer population, GATT membership status and several bilateral indicators.<sup>35</sup>  $r_i$  and  $r_j$  are the net interest margins in the source and the destination country, respectively, and  $\text{dist}_{ij}$  is the geographical distance between the two main cities of countries *i* and *j*.

#### 5.2.1. Distance effect

The regression reported in column 1 of Table 3 provides evidence that financial conditions are correlated with trade flows. Countries

<sup>35</sup> These are: common currency, regional trade agreement, agreements conferring preferential treatments (EU to ACP and ACP to EU), contiguity, common official language, common language (at least 9% spoken), colonial history, common colonizer, current colonial relationship, colonial relationship post 1945 and whether countries are or were the same country.

**Table 4**

Marginal effects of change in financing costs. This table reports the marginal effects for the regression results in Table 5. The values represent the percentage changes of exports and imports, respectively, resulting from a one percent increase in financing costs (1 + net interest margin) evaluated at the sample mean bilateral distance (8.6). Columns (1) and (2) correspond to columns (1) and (2) in Table 5. Standard errors are in parenthesis. Significance levels: \*: 10%, \*\*: 5% and \*\*\*: 1%.

| Effects from 1% increase in financing costs |                     |                     |
|---------------------------------------------|---------------------|---------------------|
| Specification                               | (1)                 | (2)                 |
| Exports                                     | −2.002***<br>(0.31) | −2.270***<br>(0.39) |
| Imports                                     | −2.280***<br>(0.30) | −1.174***<br>(0.38) |
| Mean ln dist                                | 8.595               | 8.599               |
| N                                           | 142,761             | 78,742              |

with higher net interest rate margins trade less with each other. The size of this effect is increasing in the geographical distance between trading partners. This can be seen by noting that, in line with Proposition 3, both coefficients on the distance interaction  $\zeta_3$  and  $\zeta_4$  are highly significant and negative. The preferred specification is presented in column 3, where exporter  $\times$  year and importer  $\times$  year fixed effects are included. In this specification  $\zeta_3$  and  $\zeta_4$  are larger and also highly significant.

### 5.2.2. Economic relevance

The marginal effects of financing costs evaluated at the mean log bilateral distance (8.6) for the regressions in columns 1 and 2 are reported in Table 4. They imply that a one percent increase in financing costs in a country is associated with 2.0% lower exports and 2.3% lower imports by that country. To evaluate the economic relevance of the distance interaction, consider the following comparative statics. Compare trade between Spain and Egypt (25 percentile by distance, 3355 km) with trade between Spain and South Korea (75 percentile by distance, 10,013 km). Suppose the net interest margin in Spain increased by one percent. Then we should expect Spain to have a 5.2% larger drop of its exports and a 6.3% larger drop of its imports when trading with South Korea than when trading with Egypt due to the larger geographical distance. Table 5 reports comparative statics for all specifications displayed in Table 3.

### 5.2.3. Robustness

One concern is potential omitted variable bias. If there are variables that are correlated with the net interest rate margin and bilateral

**Table 6**

Financial development, distance and export volumes. This table analyzes the relationship between financial development in the source and the destination country and export volumes. The regressions test for a direct effect of financial development and for an effect of its interaction with distance. The dependent variable is the log of exports from country  $i$  to country  $j$  in year  $t$ , 1980–2004. Financial development is proxied by private credit over GDP. Time to trade is proxied by the geographical distance between the main cities of two countries. Contract enforcement is proxied by rule of law. Regressions in columns 1 and 2 control for the log of GDP per capita, population and GATT status for exporter and importer, respectively. All regressions include a constant and control for a set of bilateral controls as discussed in the text. Column 2 also controls for contract enforcement in both countries. Errors are clustered by exporter–importer pairs. Standard errors are in parenthesis. \*, \*\* and \*\*\* denote significance at the 10%, 5% and 1% level.

| Dependent variable                      | ln bilateral exports |                     |                     |                     |
|-----------------------------------------|----------------------|---------------------|---------------------|---------------------|
|                                         | (1)                  | (2)                 | (3)                 | (4)                 |
| Exp fin devt                            | −4.465***<br>(0.29)  | −3.211***<br>(0.37) |                     |                     |
| Imp fin devt                            | −5.327***<br>(0.30)  | −2.260***<br>(0.41) |                     |                     |
| Exp fin devt $\times$ ln dist           | 0.523***<br>(0.03)   | 0.360***<br>(0.04)  | 0.552***<br>(0.03)  | 0.371***<br>(0.04)  |
| Imp fin devt $\times$ ln dist           | 0.606***<br>(0.03)   | 0.256***<br>(0.05)  | 0.618***<br>(0.04)  | 0.263***<br>(0.05)  |
| Exp law $\times$ ln dist                |                      | 0.143***<br>(0.03)  |                     | 0.156***<br>(0.03)  |
| Imp law $\times$ ln dist                |                      | 0.022<br>(0.03)     |                     | 0.009<br>(0.03)     |
| ln GDPE $\times$ ln dist                |                      | −0.095***<br>(0.03) |                     | −0.104***<br>(0.03) |
| ln GDPI $\times$ ln dist                |                      | 0.161***<br>(0.03)  |                     | 0.172***<br>(0.03)  |
| ln dist                                 | −1.981***<br>(0.03)  | −2.365***<br>(0.27) | −2.024***<br>(0.03) | −2.390***<br>(0.28) |
| R-squared                               | 0.772                | 0.786               | 0.788               | 0.792               |
| N                                       | 228,045              | 82,812              | 228,045             | 82,812              |
| # exporter–importer clusters            | 19,253               | 18,262              | 19,253              | 18,262              |
| # exporters                             | 150                  | 150                 | 150                 | 150                 |
| Country controls                        | y                    | y                   | –                   | –                   |
| Country pair controls                   | y                    | y                   | y                   | y                   |
| Importer, exporter, year FE             | y                    | y                   | –                   | –                   |
| Imp $\times$ year, exp $\times$ year FE | n                    | n                   | y                   | y                   |

trade flows that are not included in the regression, the estimate of the distance interaction can be biased. To address this issue, columns 2, 4 and 6 introduce two additional interaction terms. First, a measure of contract enforcement (rule of law) and its interaction with distance are included to control for institutional factors. Second, an interaction between the log of GDP per capita and distance is added to capture effects related to the general economic development of countries. A

**Table 5**

Comparative statics for change in financing costs. This table reports comparative statics for the regression results in Table 5. I compare trade between a country pair at the 25 distance percentile (e.g. Spain–Egypt, 3355 km) with trade between a country pair at the 75 distance percentile (e.g. Spain–South Korea, 10,013 km). Values report the reaction of trade to a one percent increase in financing costs (1 + net interest margin). Standard errors are in parenthesis. Significance levels: \*: 10%, \*\*: 5% and \*\*\*: 1%.

| Effects from 1% increase in financing costs |                     |                     |                     |                     |                     |                     |
|---------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Specification                               | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| <i>Exports</i>                              |                     |                     |                     |                     |                     |                     |
| 25 distance                                 | 0.278               | −1.468***           | –                   | –                   | –                   | –                   |
| Percentile                                  | (0.33)              | (0.44)              | –                   | –                   | –                   | –                   |
| 75 distance                                 | −4.951***           | −3.293***           | –                   | –                   | –                   | –                   |
| Percentile                                  | (0.40)              | (0.48)              | –                   | –                   | –                   | –                   |
| Difference                                  | −5.230***<br>(0.38) | −1.826***<br>(0.51) | −5.730***<br>(0.40) | −2.347***<br>(0.54) | −0.512**<br>(0.23)  | −0.089<br>(0.40)    |
| <i>Imports</i>                              |                     |                     |                     |                     |                     |                     |
| 25 distance                                 | 0.463               | 0.102               | –                   | –                   | –                   | –                   |
| Percentile                                  | (0.33)              | (0.42)              | –                   | –                   | –                   | –                   |
| 75 distance                                 | −5.827***           | −2.800***           | –                   | –                   | –                   | –                   |
| Percentile                                  | (0.39)              | (0.48)              | –                   | –                   | –                   | –                   |
| Difference                                  | −6.289***<br>(0.39) | −2.902***<br>(0.49) | −6.861***<br>(0.42) | −3.231***<br>(0.53) | −1.566***<br>(0.23) | −2.560***<br>(0.41) |
| N                                           | 142,761             | 78,742              | 142,761             | 78,742              | 142,761             | 78,742              |

comparison of Column 2 with Column 1 reveals that the introduction of these additional regressors reduces the point estimates of  $\zeta_3$  and  $\zeta_4$  to about a half of their previous values. They remain highly significant and economically relevant.

Columns 5 and 6 estimate a fixed effects model, where effects are identified from within country pair variation over time.<sup>36</sup>  $\zeta_3$  and  $\zeta_4$  become smaller but remain highly significant with the exception of  $\zeta_4$  in column 6.<sup>37</sup>

As a further robustness check, I rerun the regressions shown in Table 3 columns 1 to 4, using private credit over GDP instead of the net interest margin. The former is the standard measure for financial development, in particular, in papers that study the role of financial constraints. The results are reported in Table 6. They support the findings from the previous regressions. Note that financial development increases in the ratio of private credit over GDP. That is, the higher the ratio, the better are financial conditions. Therefore, all coefficients on the financial measure have exactly the opposite sign from the regressions in Table 3.

Can we interpret the relationship between financial conditions and trade flows as identified by the interaction terms between distance and the measures of financial conditions as causal? The main concern in this context is reverse causality. If a country does a lot of international trade, this increases its demand for financial services. A larger demand in turn can lead to efficiency gains in the provision of finance, reducing the net interest rate margin.<sup>38</sup> As discussed earlier, the distance interaction identifies effects proportional to the geographical distance between trading partners. Therefore, the relevant reverse causality to be considered is the following. Suppose there is an increase in the demand from a destination country. This increases the demand for trade finance in the source country proportional to the geographical distance from this trading partner. Reverse causality would be a problem if working capital financing for international trade was sufficiently large to have a first-order effect on the overall demand for finance in a country. While lending related to international trade finance is certainly an important activity in many countries, it can be argued that in most cases it represents a relatively small share of overall finance. A first-order effect of trade finance on the borrowing rate of firms therefore seems unlikely. This suggests that financing costs have an economically relevant effect on trade flows, proportional to distance.

## 6. Conclusions

Trade finance is a fundamental element of international trade and has become a central concern of policy makers in recent years. Yet major aspects of trade finance remain unexplored.

To shed light on the key mechanisms, this paper develops a first model of payment contract choice between Cash in Advance, Open Account and Letter of Credit, the three major contracts in international trade. The theory presented delivers several new insights. It implies that legal and financial conditions in the destination country are as relevant for trade as source country conditions. Furthermore, it shows that the payment contract choice can affect trade prices, quantities and patterns and that the ability to switch between contracts can mitigate adverse effects of financial crises on trade.

These results illustrate that the explicit modeling of trade relationships between firms in two countries can matter. In this, my paper is related to a growing literature that departs from the view of exporters that sell directly to customers in the destination market.<sup>39</sup>

<sup>36</sup> This resolves the time-invariant part of the omitted variable bias discussed in Anderson and van Wincoop (2003). An alternative would be to follow Baier and Bergstrand (2009) and explicitly introduce exogenous multilateral-resistance terms.

<sup>37</sup> This might be due to collinearity, that is the high correlation between the net interest margin and per capita GDP (−.47) and contract enforcement (−.54), respectively.

<sup>38</sup> Do and Levchenko (2007) and Braun and Raddatz (2008) find evidence for reverse causality from trade flows and trade openness, respectively, to financial development.

<sup>39</sup> See for example Araujo and Ornelas (2007), Bernard et al. (2010), Antràs and Costinot (2011) and Ahn et al. (2011).

There are several ways in which trade finance could be explored further. On the theory side, the model could be extended to allow for heterogeneity both in the firm and in the product dimension. Product differences are likely to imply different degrees of enforceability in court and different time horizons of trade relationships. Differences in firm size may affect the relative negotiation power between the exporter and the importer, the ability to enforce contracts in court and the profitability to switch between contracts in the presence of fixed costs. It would also be interesting to study the payment contract decision with exchange rate risk, which would require introducing currencies into the model.

The central challenge for further research on trade finance is to collect data on the use of different payment contracts at the firm and country pair level. While the aggregate regressions in this paper test relevant predictions of the model, more empirical work is desirable. If more detailed payment contract data was available, theoretical results on the optimal payment contract choice could be validated directly. Antràs and Foley (2011) take a first step in this direction using data from a large US food exporter. To test Corollary 1 fully, a dataset with variation across both the source and the destination country is required.

## Appendix A. Derivations of trade prices, quantities, revenues and profits

There are two cases to be considered. First, a firm can choose its optimal price and quantity independently of the other type. Alternatively, it can imitate the other type. In the following I derive both cases. To save on notation, I leave out the superscript  $c$  for the type of contract and the subscript  $i$  for the type of firm. All expressions hold for all contracts  $c \in \{CIA, OA, LC\}$  and all types  $i \in \{good, bad\}$ .

### A.1. Optimal prices and quantities – independent decision

A firm maximizes:  $\max_p E[\Pi_x] = (\alpha p^{1-\sigma} - \beta a p^{-\sigma})(P^*)^\sigma Q^*$ .

This implies:  $p_x = \frac{\beta}{\alpha} \frac{\sigma}{\sigma-1} a = \frac{\beta}{\alpha} p_d^*$ .

The expected traded quantity is:  $E[q_x] = \beta p_x^{-\sigma} (P^*)^\sigma Q^* = (\alpha)^\sigma \beta q_d^*$ .

Expected export revenues are:  $E(R_x) = p_x E[q_x] = (\alpha)^\sigma \beta R_d^*$ .

Expected profits are:  $E[\Pi_x] = \alpha p q - \beta a q = \alpha^\sigma \beta^{1-\sigma} \Pi_d^*$ .

For good firms this implies:

$$p_{x,g} = \frac{1}{\alpha} p_d^*, \quad E[q_{x,g}] = \alpha^\sigma q_d^*, \quad E(R_{x,g}) = \alpha^{\sigma-1} R_d^*, \quad E[\Pi_{x,g}] = \alpha^\sigma \Pi_d^*.$$

A bad firm not imitating a good firm would choose:

$$p_{x,b}^{nim} = \frac{\beta}{\alpha} p_d^*, \quad \text{implying} \quad E[\Pi_{x,b}^{nim}] = \alpha^\sigma \beta^{1-\sigma} \Pi_d^*.$$

### A.2. Optimal prices and quantities – bad type imitating good type

If a bad firm imitates a good firm, it chooses the same price and quantity as a good firm. Under Cash in advance, however, it only delivers the goods with probability  $\lambda$ . Its expected profits are therefore:

$$E[\Pi_{x,b}^{im}] = \alpha p^{1-\sigma} - \beta a p^{-\sigma} (P^*)^\sigma Q^* = (\alpha)^\sigma (\sigma - \beta(\sigma - 1)) \Pi_d^*.$$

## Appendix B. Introducing a contract fine

For tractability, the main analysis abstracts from any contract fine or other form of punishment imposed on a defaulting party. This section introduces a contract fine and shows how this affects the model. It derives the conditions under which this implies a voluntary

fulfillment of the contractual obligations by bad exporters and bad importers, respectively.

In the following assume that there is a contract fine that can be imposed on the defaulting party if enforcement is successful. Let this fine be exogenously given and proportional to the sales revenues  $R$  such that the total fine is  $\Delta R$ .<sup>40</sup>

The main difference implied by this additional element is that now bad firms might fulfill their obligations voluntarily. Therefore, we now have to check for an additional incentive constraint that determines whether a firm tries to default or not.

### B.1. Cash in Advance

The problem of a good exporter does not change. She always delivers the product. A bad exporter now has to decide whether to voluntarily deliver the goods or whether to try to deviate risking the contract fine.

Expected profits of a bad exporter with voluntary delivery are:

$$E[\Pi_{E,b}^{CIA,p,vd}] = C - K. \quad (33)$$

Expected profits of a bad exporter with non-voluntary delivery are:

$$E[\Pi_{E,b}^{CIA,p,nvd}] = \lambda(C - K - \Delta R) + (1 - \lambda)C = C - \lambda K - \lambda \Delta R. \quad (34)$$

Comparing these two expressions implies that non-voluntary delivery is preferred over voluntary deliver iff:

$$\frac{K}{R} > \frac{\lambda}{1 - \lambda} \Delta. \quad (35)$$

If condition (35) is violated, then the expected contract fine is sufficiently high as compared to the gains from deviating to make any firm always fulfill the contract. If the condition holds, we are back to the case analyzed in the main part of the paper.

### B.2. Open Account

Under Open Account we now have to check for the incentive constraint of a bad importer. Again, the problem of a good firm is not changed as she always pays after receipt of the goods anyways. A bad firm has to decide whether to voluntarily pay for the goods or whether to try to deviate risking the contract fine. Expected profits of a bad importer with voluntary payment are:

$$E[\Pi_{I,b}^{OA,vp}] = R - C. \quad (36)$$

Expected profits of a bad importer with non-voluntary payment are:

$$E[\Pi_{I,b}^{OA,nvp}] = R - \lambda^*(C + \Delta R). \quad (37)$$

Non-voluntary payment is therefore preferred by a bad importer iff:

$$\frac{C}{R} > \frac{\lambda^*}{1 - \lambda^*} \Delta. \quad (38)$$

If this condition is not fulfilled, all firms always follow through with their contractual obligations and the problem is solved. For a further comparison with the main text, assume in the following that this condition is fulfilled. Note that there always exists some  $\Delta$  for which this is the case. The exporter, again, has to choose between the separating and the pooling case.

### B.3. Open Account, separating case

In the separating case, profit maximization requires the participation constraint of a bad importer to bind:

$$E[\Pi_{I,b}^{OA,s,nvp}] = R - \lambda^*(C + \Delta R) = 0. \quad (39)$$

Solving for  $C$  delivers:

$$C_{I,b}^{OA,s,nvp} = \left(\frac{1}{\lambda^*} - \Delta\right)R. \quad (40)$$

There are two cases. If  $C_{I,b}^{OA,s,nvp} \leq R$ , it is optimal for the exporter to choose pooling and ask for  $C = R$ . If  $C_{I,b}^{OA,s,nvp} > R$ , the separating case might be profitable. I compare this case in the following with the pooling case to determine the condition under which this is true. Under the separating case with non-voluntary payment, noting that a share of  $1 - \eta$  firms are of bad type and accept the contract, the expected profits of an exporter are:

$$E[\Pi_E^{OA,s,nvp}] = (1 - \eta^*) \left[ \frac{\lambda^*(C - \delta R)}{(1 + r)^t} - K \right]. \quad (41)$$

Plugging in  $C$  delivers:

$$E[\Pi_E^{OA,s,nvp}] = (1 - \eta^*) \left[ \frac{1 - \lambda^*(\Delta + \delta)}{(1 + r)^t} R - K \right]. \quad (42)$$

### B.4. Open Account, pooling

Under pooling  $C^{OA,p} = R$ . This implies:

$$E[\Pi_E^{OA,p,nvp}] = \frac{\eta^* + (1 - \eta^*)\lambda^*(1 - \delta)}{(1 + r)^t} R - K. \quad (43)$$

### B.5. Choice between separating and pooling by the exporter

Combining expressions (42) and (43) an exporter prefers pooling iff:

$$\frac{R}{K} > \frac{\eta^*(1 + r)^t}{\eta^* - (1 - \eta^*)[(1 - \lambda^*(1 + \Delta))]} \quad (44)$$

Note that this condition simplifies to Eq. (20) in the main text for  $\Delta = 0$ .

### B.6. Letter of Credit

Under a Letter of Credit, deviating is never optimal as an exporter only gets paid after delivery and an importer only receives the goods

<sup>40</sup> The cost of breaking the contract could also be driven by a loss in future gains from trade. In Schmidt-Eisenlohr (2011), I study this explicitly by looking at repeated contracts.



after payment. Contract fines therefore do not change the problem of the firms.

### B.7. Summary

When a contract fine is added to the model, additional incentive constraints have to be taken into account. That is, if the contract fine is sufficiently large both firms always fulfill their contract voluntarily. If the contract fine is too small, the commitment problem remains and the mechanism discussed in the main part of the paper prevails.

## Appendix C. Importer bargaining power

In this section, Cash in Advance, Open Account and Letter of Credit are analyzed under the alternative assumption that all bargaining power lies with the importer.

### C.1. Cash in Advance – overview

The importer has two choices. She can pay an amount in advance that is only accepted by bad exporters (separating case) or she can pay a sufficiently high amount in advance to also make good exporters accept the offer (pooling). In the following I solve the problem for both cases and compare the outcomes. Then, similarly to the analysis in the main text, I derive a condition under which only pooling occurs.

### C.2. Cash in Advance – separating case

The participation constraint of a bad exporter is:

$$E[\Pi_{E,b}^{CIA,s}] = C^{CIA,s} - \lambda K \geq 0. \quad (45)$$

A binding constraint therefore implies:  $C^{CIA,s} = \lambda K$ . This leads to the following expected profits of an importer under Cash in Advance, noting that only a fraction  $1 - \eta$  of exporters accept this contract:

$$E[\Pi_I^{CIA,s}] = (1 - \eta) \left( \frac{\lambda(1 - \delta)}{(1 + r^*)^t} R - \lambda K \right). \quad (46)$$

### C.3. Cash in Advance – pooling case

The participation constraint of a good exporter is:

$$E[\Pi_{E,g}^{CIA,p}] = C^{CIA,p} - K \geq 0. \quad (47)$$

A binding constraint therefore implies:  $C^{CIA,p} = K$ . Leading to the following expected profits for an importer:

$$E[\Pi_I^{CIA,p}] = \frac{\eta + (1 - \eta)\lambda(1 - \delta)}{(1 + r^*)^t} R - K. \quad (48)$$

Comparing profits from the pooling and the separating case it is easy to see that pooling is preferred by the importer if:

$$\frac{R}{K} > (1 - (1 - \eta)\lambda) \frac{(1 + r^*)^t}{\eta}. \quad (49)$$

### C.4. Open Account

Under Open Account the separating case is never chosen by the importer. A bad importer would have to pay more to signal her type, which can never be optimal. A good importer cannot signal her type by paying

less as this can always be mimicked by the bad type. The participation constraint of a good exporter under pooling is:

$$E[\Pi_{E,g}^{OA,p}] = \frac{\eta^* + (1 - \eta^*)\lambda^*(1 - \delta)}{(1 + r)^t} C^{OA,p} - K \geq 0. \quad (50)$$

A binding constraint therefore implies that  $C^{OA,p} = \frac{(1 + r)^t}{\eta^* + (1 - \eta^*)\lambda^*(1 - \delta)} K$ . This leads to the following expected profits of a good importer:

$$E[\Pi_{I,g}^{OA,p}] = R - \frac{(1 + r)^t}{\tilde{\lambda}^*} K. \quad (51)$$

The expected profits of a bad importer are:

$$E[\Pi_{I,b}^{OA,p}] = R - \lambda^* \frac{(1 + r)^t}{\tilde{\lambda}^*} K. \quad (52)$$

### C.5. Letter of Credit

Under a Letter of Credit the exporter participation constraint is:

$$E[\Pi_E^{LC}] = \frac{C^{LC}}{(1 + r)^t} - K \geq 0. \quad (53)$$

A binding constraint implies that  $C^{LC} = K(1 + r)^t$ . This leads to the following expected profits by the importer:

$$E[\Pi_I^{LC}] = \frac{1}{(1 + r^*)^t} R - \left( \frac{1}{(1 + r^*)^t} + f^{LC} \right) (1 + r)^t K. \quad (54)$$

### C.6. Payment contract comparisons

Comparing the equations on expected importer profits (48), (52) and (54) with the equations on the expected exporter profits in the main text (6), (16) and (23) reveals the same patterns as derived formally for Proposition 1 and for Corollary 1. In particular, the effects of financing costs and enforcement in the source and the destination country have the same directions on the payment contract choice, independently whether the exporter or the importer has all bargaining power.

## Appendix D. Testing the mechanism

The empirical section of the paper tested for the effects of source and destination country financing conditions on aggregate trade flows. In this section, I derive a test that is more closely related to the payment contract choice as modeled in this paper. In general, as stated in Proposition 1, the choice between exporter finance (OA) and importer finance (CIA) depends both on contract enforcement and interest rates in the source and the destination country. Comparing enforcement and interest rates of two countries and focusing on the choice between CIA and OA, four cases are possible: In cases I and IV, there is a clear prediction on the payment contract, which is independent of the relative effect of enforcement and financing costs. In both cases, one country has an absolute advantage in financing costs and the other country has an absolute advantage in contract enforcement. The model predicts that in case I, CIA is chosen and thus only the destination country interest rate should matter for trade. In case IV, OA is preferable and thus only the source country interest rate should affect trade volumes.

| CIA vs. OA: four cases                   |                    |                   |
|------------------------------------------|--------------------|-------------------|
|                                          | $r > r^*$          | $r \leq r^*$      |
| $\tilde{\lambda} > \tilde{\lambda}^*$    | I: always CIA      | II: both possible |
| $\tilde{\lambda} \leq \tilde{\lambda}^*$ | III: both possible | IV: always OA     |

**Table 7**

Minimum financing costs, distance and exports. This table analyzes the effect of the minimum of financing costs of the exporting and importing country and its interactions with distance on export volumes. The dependent variable is the log of exports from country  $i$  to country  $j$  in year  $t$ , 1987–2004. The sample is restricted to cases where  $r > r^*$  &  $\lambda > \lambda^*$  or  $r \leq r^*$  &  $\lambda \leq \lambda^*$ . Financing costs are measured by the net interest margin. Time to trade is proxied by the geographical distance between the main cities of two countries. Contract enforcement is proxied by Rule of Law. Regressions in columns 1 to 3 control for the log of GDP per capita, population and GATT status for exporter and importer, respectively. All regressions include a constant and a set of bilateral controls as discussed in the text. Columns 2 and 3 also control for contract enforcement in both countries. Errors are clustered by exporter–importer pairs. Standard errors are in parenthesis. Significance levels: \*: 10%, \*\*: 5% and \*\*\*: 1%.

| Dependent variable                            | Ln bilateral exports |                           |                             |                      |                           |                             |
|-----------------------------------------------|----------------------|---------------------------|-----------------------------|----------------------|---------------------------|-----------------------------|
| Specification                                 | (1)                  | (2)                       | (3)                         | (4)                  | (5)                       | (6)                         |
| Ln exp int                                    | 26.717***<br>(7.77)  | – 1.415<br>(1.29)         | 7.910<br>(9.88)             |                      |                           |                             |
| Ln imp int                                    | 19.455**<br>(7.99)   | – 3.545***<br>(1.30)      | 0.612<br>(10.45)            |                      |                           |                             |
| Ln exp int $\times$ Ln dist                   | – 3.406***<br>(0.90) |                           | – 1.046<br>(1.15)           | – 3.609***<br>(0.98) |                           | – 1.094<br>(1.27)           |
| Ln imp int $\times$ Ln dist                   | – 2.856***<br>(0.94) |                           | – 0.481<br>(1.22)           | – 2.995***<br>(1.01) |                           | – 0.465<br>(1.33)           |
| <b>Ln min int</b>                             |                      | <b>– 3.917*</b><br>(2.15) | <b>52.051***</b><br>(18.58) |                      | <b>– 5.016*</b><br>(2.86) | <b>53.405***</b><br>(19.93) |
| <b>Ln min int <math>\times</math> Ln dist</b> |                      |                           | <b>– 6.702***</b><br>(2.17) |                      |                           | <b>– 7.037***</b><br>(2.31) |
| Law of min int                                |                      | – 0.255***<br>(0.08)      | 1.789**<br>(0.81)           |                      | – 0.264***<br>(0.10)      | 2.045**<br>(0.84)           |
| Law of min int $\times$ Ln dist               |                      |                           | – 0.227**<br>(0.09)         |                      |                           | – 0.258***<br>(0.10)        |
| Ln GDP of min int                             |                      | 0.061<br>(0.06)           | – 1.188**<br>(0.53)         |                      | 0.056<br>(0.06)           | – 1.159**<br>(0.56)         |
| Ln GDP of min int $\times$ Ln dist            |                      |                           | 0.143**<br>(0.06)           |                      |                           | 0.140**<br>(0.06)           |
| Ln dist                                       | – 2.471***<br>(0.41) | – 1.354***<br>(0.04)      | – 2.513***<br>(0.41)        | – 2.485***<br>(0.42) | – 1.350***<br>(0.04)      | – 2.557***<br>(0.42)        |
| Exp law $\times$ Ln dist                      | 0.116**<br>(0.05)    |                           | 0.229***<br>(0.06)          | 0.107**<br>(0.05)    |                           | 0.236***<br>(0.06)          |
| Imp law $\times$ Ln dist                      | 0.051<br>(0.05)      |                           | 0.165**<br>(0.07)           | 0.051<br>(0.05)      |                           | 0.182**<br>(0.08)           |
| Ln GDPE $\times$ Ln dist                      | – 0.101**<br>(0.04)  |                           | – 0.170***<br>(0.05)        | – 0.103**<br>(0.04)  |                           | – 0.170***<br>(0.05)        |
| Ln GDPI $\times$ Ln dist                      | 0.259***<br>(0.04)   |                           | 0.188***<br>(0.05)          | 0.266***<br>(0.04)   |                           | 0.197***<br>(0.05)          |
| R-squared                                     | 0.810                | 0.804                     | 0.810                       | 0.824                | 0.819                     | 0.825                       |
| N                                             | 21,119               | 21,119                    | 21,119                      | 21,119               | 21,119                    | 21,119                      |
| # exporter–importer clusters                  | 8015                 | 8015                      | 8015                        | 8015                 | 8015                      | 8015                        |
| # exporters                                   | 144                  | 144                       | 144                         | 144                  | 144                       | 144                         |
| Country controls                              | y                    | y                         | –                           | –                    | –                         | –                           |
| Country pair controls                         | y                    | y                         | y                           | y                    | y                         | y                           |
| Importer, exporter, year FE                   | y                    | y                         | y                           | –                    | –                         | –                           |
| Imp $\times$ year, exp $\times$ year FE       | n                    | n                         | n                           | y                    | y                         | y                           |

**Proposition 4.** Suppose firms choose between CIA and OA and suppose  $\tilde{\lambda} > \tilde{\lambda}^*$  and  $r > r^*$  or  $\tilde{\lambda} \leq \tilde{\lambda}^*$  and  $r \leq r^*$ . Then, the log of expected export revenues

- i) decreases in the log of the minimum interest rate:  $\frac{\partial E[\ln R_{ij}]}{\partial \ln\{ \ln(1+r_i), \ln(1+r_j) \}} < 0$ .  
ii) the more so, the larger  $\ln t$   $\frac{\partial^2 E[\ln R_{ij}]}{\partial \ln\{ \ln(1+r_i), \ln(1+r_j) \} \partial \ln t} < 0$ .

**Proof.** See Appendix F.  $\square$

If one country has an absolute advantage in financing ( $r_i < r_j$ ) and the other country has an absolute advantage in fulfilling contracts ( $\tilde{\lambda}_j > \tilde{\lambda}_i$ ), then the payment contract is clearly determined. That is the choice is independent of the relative importance of enforcement as compared to financing costs. In this case, only the minimum financing cost of the two countries matters. This effect increases in the log of the time needed for trade  $\ln t$ .

#### D.1. Testing the mechanism

To test Proposition 4 empirically, I restrict the sample to include all observations where either I or IV is the case, measuring  $r$  by the net interest rate margin and  $\tilde{\lambda}$  by rule of law. This reduces the sample size from 78,742 to 21,119. As can be seen by comparing Table 2a with Table 2c, the summary statistics of this subset are very similar

to those of the full sample. Note, in particular, as reported in Table 7, that the number of exporters does not change. That is, any country has some country with which as a pair if fulfills the condition. In 2004, at the low income end an example of such a country pair is Burundi and Nigeria, in the middle income group it is Hungary and Vietnam and in the high income group it is Norway and Germany. The baseline specification (Table 7, column 3) for the test on the minimum financing cost is:

$$\begin{aligned} \ln Y_{ijt} = & \zeta_0 + \zeta_1 \ln(1+r_i) + \zeta_2 \ln(1+r_j) + \zeta_3 \ln(\text{dist}_{ij}) \cdot \ln(1+r_i) \\ & + \zeta_4 \ln(\text{dist}_{ij}) \cdot \ln(1+r_j) + \zeta_5 \ln(\min\{1+r_i, 1+r_j\}) \\ & + \zeta_6 \ln(\text{dist}_{ij}) \cdot \ln(\min\{1+r_i, 1+r_j\}) + \zeta_7 \ln(\text{dist}_{ij}) \\ & + \theta'X + \chi_i + \chi_j + \chi_t + \epsilon_{ijt}. \end{aligned} \quad (55)$$

The main prediction from Proposition 4 is first, that the minimum interest rate has a negative effect on the volume of trade ( $\zeta_5 + \zeta_6 \sum_1^N (\ln \text{dist}_{ij}) < 0$ ) and second, that this effect is increasing in the log of time needed for trade  $\ln t$  ( $\zeta_6 < 0$ ). Furthermore, controlling for the minimum interest rate and its interaction with distance, the source and the destination country interest rates and their interactions with distance should not affect trade. That is  $\zeta_1, \zeta_2, \zeta_3, \zeta_4 = 0$ .

Additionally, to ensure that the minimum net interest rate margin is not picking up an effect of GDP per capita or contract enforcement, columns 2, 3, 5 and 6 control for the contract enforcement and the GDP per capita in the country with the minimum interest rate.<sup>41</sup> Columns 3 and 6 furthermore control for the interactions between distance and a set of controls. These are the enforcement (rule of law) and the GDP per capita in the country with the minimum interest rate, GDP per capita in the source and destination country and enforcement (rule of law) in the source and the destination country.<sup>42</sup>

First, I rerun the regressions from Table 3 columns 2 and 4 with the new sample, which is reported in Table 7, columns 1 and 4. I find somewhat larger coefficients on the interaction between distance and financing costs in the source and destination country, but overall the results are very similar to those in Table 3. Columns 2 and 4 test for the direct effect of the minimum interest rate.  $\zeta_5$  is negative in both cases and significant at the 10% level. Columns 3 and 6 test for the interaction between distance and the minimum interest rate. The coefficients for  $\zeta_6$  are large, negative and highly significant. The average effect is negative and significant. Furthermore, as predicted by the theory, coefficients  $\zeta_1, \zeta_2, \zeta_3$  and  $\zeta_4$  all become insignificant. That is, after controlling for the minimum financing costs, the source and destination country financing costs do not matter. This is evidence for the hypothesis that if one country has an absolute advantage in financing whereas the other country has an absolute advantage in enforcement, trade transactions are financed by the side with access to cheaper funds. Note that, given that each country is required to have an absolute advantage, this exercise focuses on country pairs that are not too dissimilar.<sup>43</sup> An alternative explanation for this empirical finding, that does not rely on the trade-off derived in the payment contract choice model, is thus difficult to come up with.

## Appendix E. The case of $\eta = 0$ and $\eta^* = 0$

This section illustrates the simpler model as introduced in Schmidt-Eisenlohr (2009) where there are no good firms, that is  $\eta = 0$  and  $\eta^* = 0$  and where the cost of enforcing contracts is zero  $\delta = 0$ . This model has been adapted among others by Antràs and Foley (2011) and Glady and Potin (2011).

In order to make the payment choice contract model work, an additional assumption which I call the limited value of contract constraint has to be made. This assumption requires that the payment value agreed upon does not exceed the sales value of the goods in the destination market. Now, the exporter maximizes her expected profits taking into account the participation constraint of the importer and the limited value of contract constraint.

### E.1. Cash in Advance

The maximization problem now is:

$$\max_c E[\Pi_E^{CIA}] = C^{CIA} - \lambda K, \quad (56)$$

$$\text{s.t. } C^{CIA} \leq R, \quad (\text{limited value of contract}) \quad (57)$$

<sup>41</sup> Enforcement and GDP per capita are strongly negatively correlated with the net interest rate margin (−.54 and −.47 respectively).

<sup>42</sup> In an alternative robust check I control for the maximum enforcement of each country pair and the maximum GDP per capita of each country pair and their interactions with distance. Results do not change under this alternative specification. They are available upon request.

<sup>43</sup> As mentioned above, enforcement and the net interest rate margin are strongly negatively correlated. Therefore, on average countries with the better enforcement also have the lower financing costs.

$$\text{and } E[\Pi_I^{CIA}] = \lambda R - (1 + r^*)^t C^{CIA} \geq 0. \quad (58)$$

(participation constraint importer).

Under Cash in Advance, the participation constraint of the importer always binds. This implies the following optimal payment and expected profits:

$$C^{CIA} = \frac{\lambda}{(1 + r^*)^t} R x m, \quad E[\Pi_E^{CIA}] = \frac{\lambda}{(1 + r^*)^t} R - \lambda K. \quad (59)$$

### E.2. Open Account

The exporter now maximizes the following problem:

$$\max_c E[\Pi_E^{OA}] = \frac{1}{(1 + r)^t} (\lambda^* C^{OA} - K(1 + r)^t), \quad (60)$$

$$\text{s.t. } C^{OA} \leq R, \quad (\text{limited value of contract}) \quad (61)$$

$$\text{and } E[\Pi_I^{OA}] = \frac{1}{(1 + r^*)^t} (R - \lambda^* C^{OA}) \geq 0, \quad (62)$$

(participation constraint importer),

Under Open Account, the limited value of contract condition always binds. The optimal payment amount and expected profits are thus:

$$C^{OA} = R, \quad E[\Pi_E^{OA}] = \frac{\lambda^*}{(1 + r)^t} R - K. \quad (63)$$

### E.3. Letter of Credit

As a Letter of Credit fully resolves the enforcement problem, its analysis is independent of the share of good firms  $\eta$ . For more details on the simple model please see Schmidt-Eisenlohr (2009).

## Appendix F. Proofs

**Proof of Proposition 1.** Given the assumptions on parameters, only pooling contracts are offered in equilibrium. Thus, CIA is only used if this payment contract maximizes expected profits of good exporters. For OA and LC expected exporter profits are equal for both types. Therefore, the expected profits of good exporters completely determine the payment contract choice. Expressions in Proposition 1 thus follow directly from combining Eqs. (6), (16) and (23).

**Proof of Corollary 1.** The three conditions are:

$$I: \text{OA preferred to CIA} \Leftrightarrow \frac{\tilde{\lambda}^*}{(1 + r)^t} - \frac{\tilde{\lambda}}{(1 + r^*)^t} > 0,$$

$$II: \text{OA preferred to LC} \Leftrightarrow \frac{1}{(1 + r)^t} \left[ \tilde{\lambda}^* - \frac{1}{1 + f^{LC}(1 + r^*)^t} \right] > 0,$$

$$III: \text{CIA preferred to LC} \Leftrightarrow \frac{\tilde{\lambda}}{(1 + r^*)^t} - \frac{1}{(1 + r)^t (1 + f^{LC}(1 + r^*)^t)} > 0.$$

Define  $\kappa = \frac{\tilde{\lambda}(1 + r)^t [1 + f^{LC}(1 + r^*)^t]^2}{(1 + r^*)^{2t}}$ . The following table shows

whether the sign of each of the three conditions I–III is more likely to be positive or negative if a parameter  $\in \{r, r^*, \lambda, \lambda^*, \eta, \eta^*, f^{LC}, \delta\}$  changes. + indicates that the condition is more likely to hold if the parameter increases. − indicates that the condition is less likely to hold if the parameter increases. 0 indicates that a change in the parameter has no effect on whether the condition holds.

### F.1. Effects of changes in financing and contracting conditions on contract choice

This table reports how the contract choice is affected by changes in the four variables characterizing financing conditions and enforcement probabilities and costs. Each row refers to one of the four variables. Each column refers to one of the three conditions from Proposition 1 comparing two contracts types. The signs indicate whether an increase in the variable makes it more or less likely that the first payment contract is chosen over the second. In column 1 for example, the first minus sign indicates that a higher  $r$  makes OA less and CIA more attractive.

| Condition variable | I (OA vs. CIA) | II (OA vs. LC) | III (CIA vs. LC)        |
|--------------------|----------------|----------------|-------------------------|
| $r$                | —              | 0              | +                       |
| $r^*$              | +              | +              | —                       |
|                    |                |                | (if $f^{LC} < \kappa$ ) |
| $\lambda$          | —              | 0              | +                       |
| $\lambda^*$        | +              | +              | 0                       |
| $\eta$             | —              | 0              | +                       |
| $\eta^*$           | +              | +              | 0                       |
| $f^{LC}$           | 0              | +              | +                       |
| $\delta$           | amb.           | —              | —                       |

Combining the results from the above table implies: i) CIA weakly increases in  $r$ ,  $\lambda$ ,  $\eta$  and  $f^{LC}$  and weakly decreases in  $\lambda^*$  and  $\eta^*$ . It also weakly decreases in  $r^*$  if  $f^{LC} < \kappa$ ; ii) OA weakly increases in  $r^*$ ,  $\lambda^*$ ,  $\eta^*$  and  $f^{LC}$  and weakly decreases in  $r$ ,  $\lambda$  and  $\eta$ ; iii) LC weakly increases in  $\delta$  and weakly decreases in  $r$ ,  $\lambda$ ,  $\lambda^*$ ,  $\eta$ ,  $\eta^*$  and  $f^{LC}$ .

**Proof of Proposition 2.** From before, expected revenues and profits for a good firm are (as  $\beta = 1$ ):  $E[R_X] = (\alpha^c)^{\sigma-1} R_d^*$  and  $E[\Pi_X] = (\alpha^c)^{\sigma} \Pi_d^*$ .

Note that:  $E[\Pi_X^{1,g}] \geq E[\Pi_X^{2,g}] \Leftrightarrow E[R_X^{1,g}] \geq E[R_X^{2,g}]$ . That is, for a good exporter, the payment contract which maximizes expected profits also implies the highest expected revenues. If a good exporter is indifferent between two payment contracts, they deliver the same expected profits and imply the same expected revenues.

There are two cases which can trigger a payment contract change:

- 1) Financing costs increase or the enforcement probability decreases in one country, implying lower profits. Then, the contract is only changed if this limits the reduction in profits and revenues. This cannot lead to a net increase in profits and revenues, because this would be a contradiction to profit maximization in the first place.
- 2) Financing costs decrease or the enforcement probability increases in one country, implying higher profits. Then, the contract is only changed if this implies even higher profits and revenues. This cannot lead to a net reduction in profits and revenues, because this would be a contradiction to profit maximization in the first place.

Therefore, a change in payment contract does not change the sign of the derivative of the expected revenues with respect to the parameters of interest ( $r, r^*, \lambda, \lambda^*$ ).

Next, I prove the statements with respect to the specific parameters. In general, expected revenues are:

$$E[R_{X,i}^c] = p_X E[q_X] = \frac{1}{\alpha^c} A^{c,i} R_d^* = (\alpha^c)^{\sigma-1} \beta^{c,i} R_d^*.$$

With:

$$\alpha^{CIA} = \tilde{\lambda} (1+r)^{-t}, \alpha^{OA} = \tilde{\lambda}^* (1+r)^{-t},$$

$$\alpha^{LC} = (1+r)^{-t} (1+f^{LC} (1+r^*)^t)^{-1}.$$

And:

$$\beta^{CIA,g} = \beta^{OA} = \beta^{LC} = 1, \beta^{CIA,b} = \lambda.$$

As  $\sigma > 1$ :  $\frac{\partial E[R_X]}{\partial \alpha^c} > 0$ . Also note that:  $\frac{\partial E[R_X]}{\partial \beta^{c,i}} > 0$ .

Proof part 1 (for all firms):

(i)

$$\frac{\partial E[R_{X,i}^c]}{\partial (1+r)} \leq 0 \text{ as } \frac{\partial \alpha^c}{\partial (1+r)} \leq 0 \text{ and } \frac{\partial \beta^{c,i}}{\partial (1+r)} = 0$$

$$\forall c \in \{CIA, OA, LC\} \text{ and } \forall i \in \{good, bad\}.$$

$$\frac{\partial E[R_{X,i}^c]}{\partial (1+r^*)} \leq 0 \text{ as } \frac{\partial \alpha^c}{\partial (1+r^*)} \leq 0 \text{ and } \frac{\partial \beta^{c,i}}{\partial (1+r^*)} = 0$$

$$\forall c \in \{CIA, OA, LC\} \text{ and } \forall i \in \{good, bad\}.$$

$$\frac{\partial E[R_{X,i}^c]}{\partial (1+r)} + \frac{\partial E[R_X]}{\partial (1+r^*)} < 0 \text{ as either } \frac{\partial \alpha^c}{\partial (1+r)} < 0$$

$$\text{or } \frac{\partial \alpha^c}{\partial (1+r^*)} < 0 \quad \forall c \in \{CIA, OA, LC\}$$

$$\text{and } \frac{\partial \beta^{c,i}}{\partial (1+r)} = 0 \quad \forall c \in \{CIA, OA, LC\} \text{ and } \forall i \in \{good, bad\}.$$

(ii)

$$\frac{\partial E[R_{X,i}^c]}{\partial \lambda^*} \geq 0 \text{ as } \frac{\partial \alpha^c}{\partial \lambda^*} \geq 0 \text{ and } \frac{\partial \beta^{c,i}}{\partial \lambda^*} = 0$$

$$\forall c \in \{CIA, OA, LC\} \text{ and } \forall i \in \{good, bad\}.$$

$$\frac{\partial E[R_{X,i}^c]}{\partial \lambda} \geq 0 \text{ as } \frac{\partial \alpha^c}{\partial \lambda} \geq 0 \text{ and } \frac{\partial \beta^{c,i}}{\partial \lambda} \geq 0$$

$$\forall c \in \{CIA, OA, LC\} \text{ and } \forall i \in \{good, bad\}.$$

**Proof of Proposition 3.** As in Proposition 2, the result holds for good exporters when allowing for an endogenous change of payment contract. The proof for this is analogous to the proof above. Next, to prove the statements with respect to the specific parameters, note that:  $\ln(E[R_{X,i}^c]) = (\sigma-1) \ln \alpha^c + \ln \beta^{c,i} + \ln R_d^*$ . Now,  $\ln \alpha^{CIA} = \ln \tilde{\lambda} - t \ln(1+r)$ ,  $\ln \alpha^{OA} = \ln \tilde{\lambda}^* - t \ln(1+r)$  and  $\ln \alpha^{LC} = -t \ln(1+r) - \ln(1+f^{LC}(1+r^*)^t)$ . Taking the derivatives with respect to  $\ln(1+r)$  and  $\ln(1+r^*)$ , and the two cross derivatives with  $\ln t$  it is easy to check the results stated in the Proposition.

**Proof of Proposition 4.** Revenues are given by:  $\ln E[R_{X,i}^c] = (\sigma-1) \ln \alpha^c + \ln \beta^{c,i} + \ln R_d^*$

i) There are two cases:

- 1) if  $\tilde{\lambda} > \tilde{\lambda}^*$  and  $r > r^*$ , then CIA is optimal. Then,  $\ln \alpha = \ln \alpha^{CIA} = \ln \tilde{\lambda} - t \ln(1+r)$  and  $\frac{\partial \ln \beta}{\partial \ln(1+r)} = \frac{\partial \ln \beta}{\partial \ln(1+r^*)} = 0$ . As  $\frac{\partial \alpha}{\partial (1+r)} < 0$  and  $\frac{\partial \alpha}{\partial (1+r^*)} = 0$ , the log of expected revenues decreases in  $\ln(1+r)$  and is independent of  $\ln(1+r^*)$ .
- 2) if  $\tilde{\lambda} \leq \tilde{\lambda}^*$  and  $r \leq r^*$ , then OA is optimal. Then,  $\ln \alpha = \ln \alpha^{OA} = \ln \tilde{\lambda}^* - t \ln(1+r)$  and  $\frac{\partial \ln \beta}{\partial \ln(1+r)} = \frac{\partial \ln \beta}{\partial \ln(1+r^*)} = 0$ . As  $\frac{\partial \ln \alpha}{\partial \ln(1+r)} < 0$  and  $\frac{\partial \ln \alpha}{\partial \ln(1+r^*)} = 0$ , the log of expected revenues decreases in  $\ln(1+r)$  and is independent of  $\ln(1+r^*)$ .

ii) Follows from i) and taking the cross derivatives with respect to the log of the respective interest rates and the log of distance.



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